THEORY

Training Program 01

Advanced NLP & Large Language Models (LLMs)

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Training Program 01

- **M01:** Text Preprocessing
- **M02:** Word and Sentence Embeddings
- **M03:** Transformer Architectures
- **M04:** Entity-Centric NLP
- **M05: Applied NLP Tasks**
- **M06:** Multimodal NLP
- **M07:** LLM Fundamentals
- **M08:** Prompt Engineering
- **M09:** Fine-Tuning & PEFT
- **M10:** RAG & Real-World Projects

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MODULE M01

TEXT PREPROCESSING

CHAPTER M05

Applied NLP Tasks

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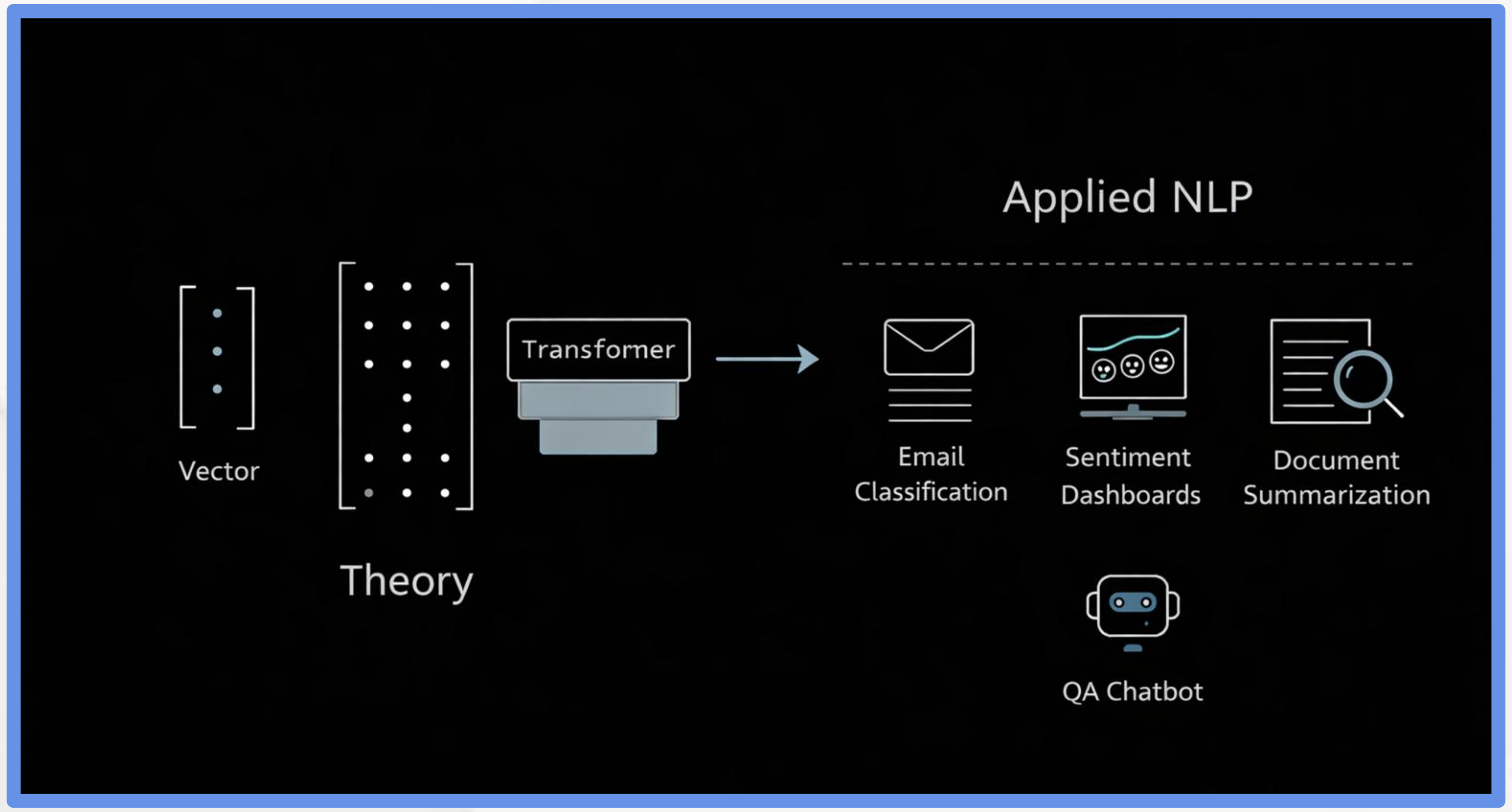
Chapter 5.1 Introduction to Applied NLP Tasks

From Theory to Applied NLP

- NLP moves from mathematical modeling to real-world problem solving.
- Applied NLP focuses on business, industrial, and societal value.
- Models are evaluated by usefulness, not only benchmark scores.
- This module connects Transformers to production systems.



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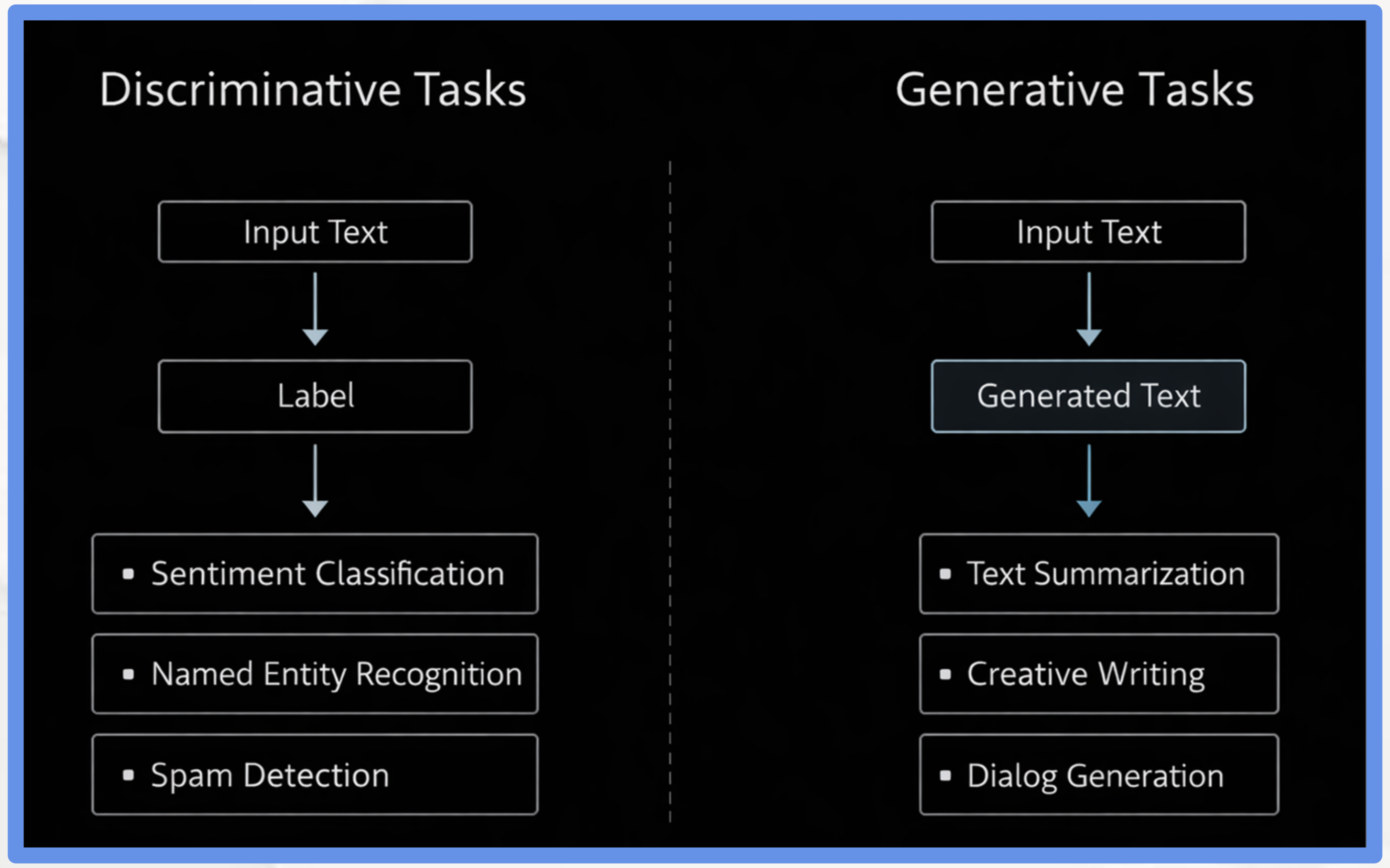
Chapter 5.1 Introduction to Applied NLP Tasks

Applied NLP Task Families

- Applied NLP tasks fall into two paradigms
- Tasks are defined by what the model produces
- Choice impacts architecture, evaluation, and cost



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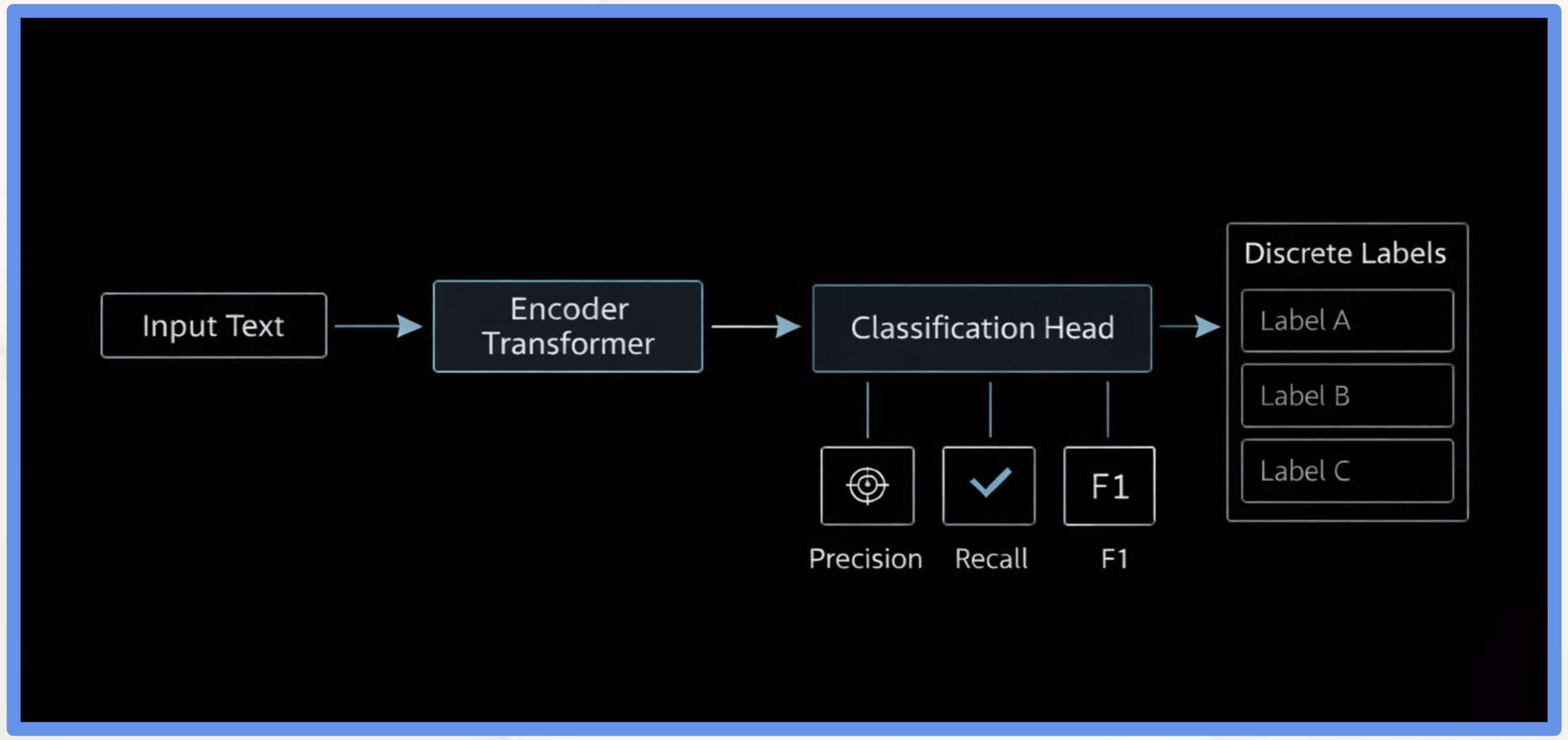
Chapter 5.1 Introduction to Applied NLP Tasks

Discriminative Tasks

- Output: predefined labels
- Goal: decision, filtering, categorization
- Typical architectures: Encoder-only (BERT-style)
- Metrics: Accuracy, Precision, Recall, F1-score
- Examples: classification, sentiment, NER



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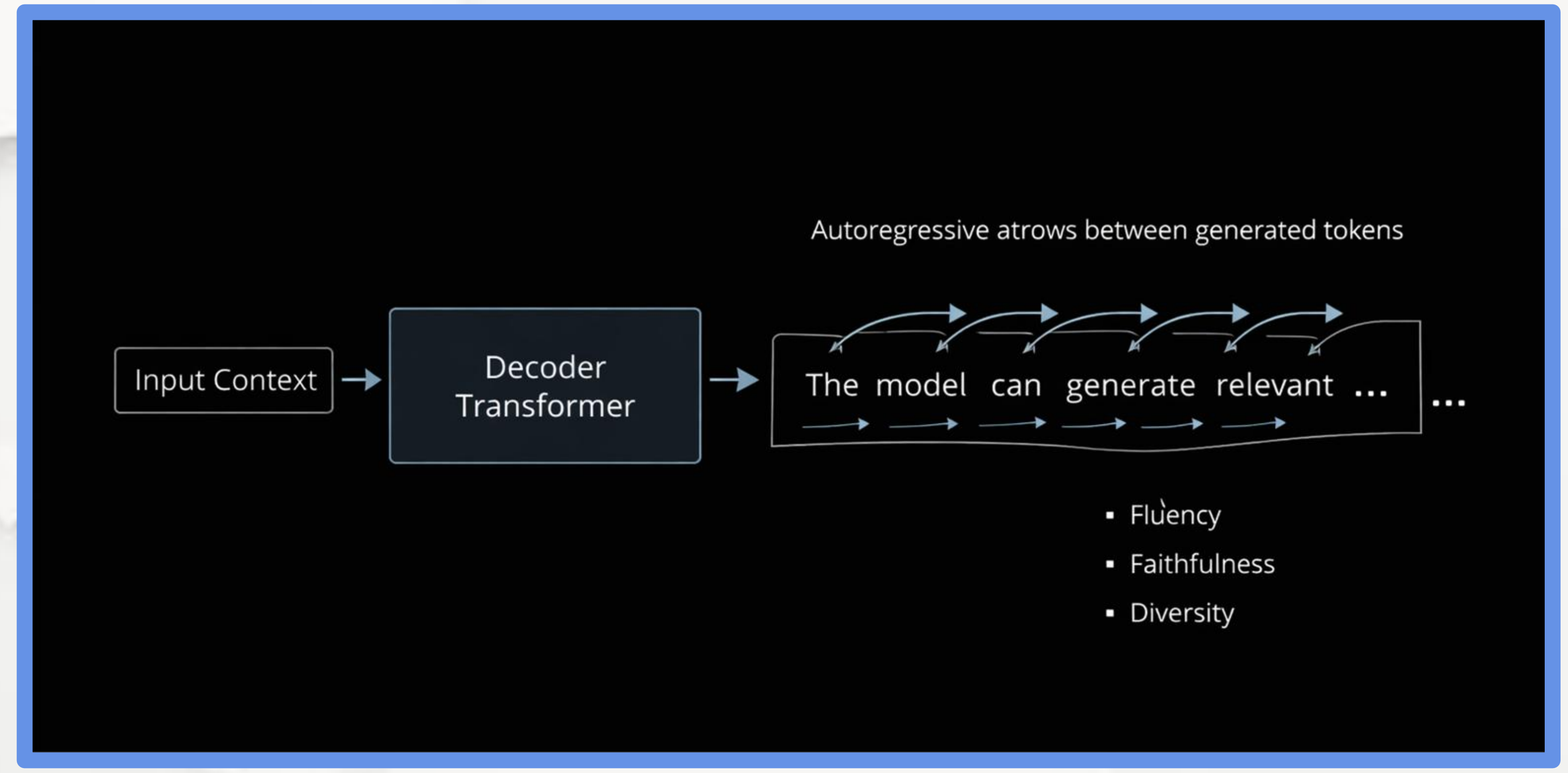
Chapter 5.1 Introduction to Applied NLP Tasks

Generative Tasks

- Output: newly generated text
- Goal: synthesis, explanation, abstraction
- Typical architectures: Decoder or Encoder–Decoder
- Evaluation is probabilistic and semantic
- Examples: summarization, QA, text generation



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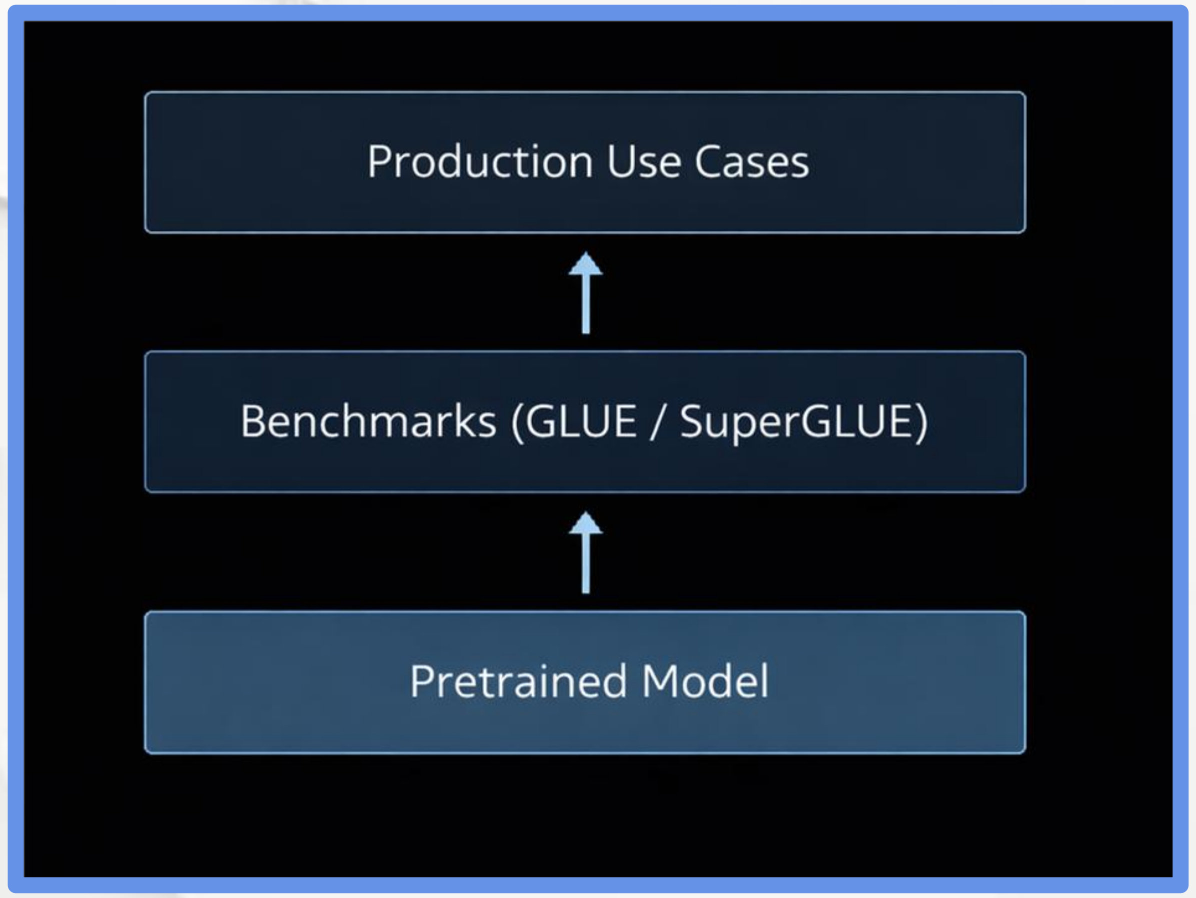
Chapter 5.1 Introduction to Applied NLP Tasks

Benchmarks and Model Readiness

- Benchmarks standardize model comparison
- GLUE and SuperGLUE measure reasoning ability
- High benchmark scores ≠ production readiness
- Fine-tuning adapts models to domain-specific tasks



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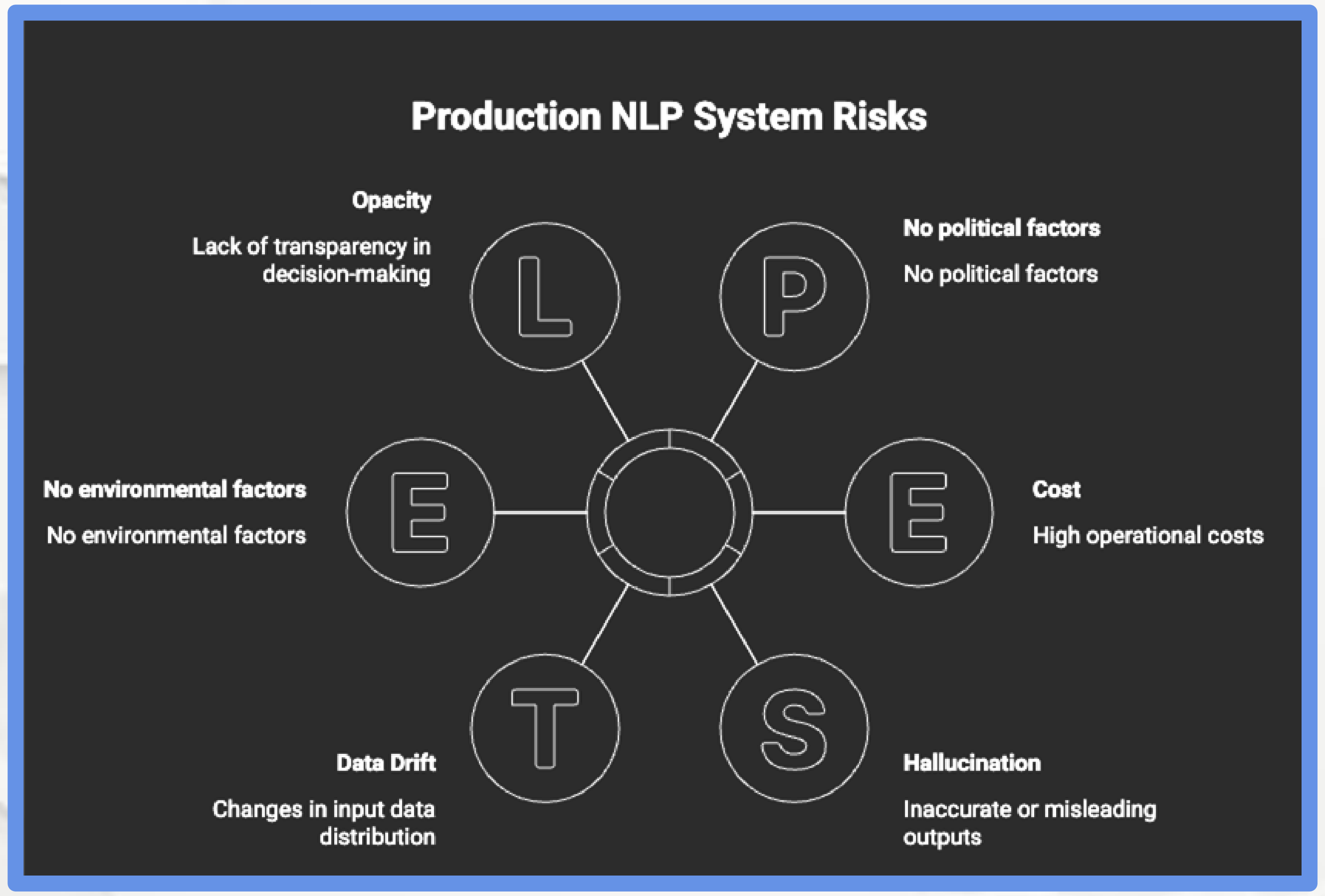
Chapter 5.1 Introduction to Applied NLP Tasks

Production Challenges in Applied NLP

- Benchmarks standardize model comparison.
- GLUE and SuperGLUE measure reasoning ability.
- High benchmark scores ≠ production readiness.
- Fine-tuning adapts models to domain-specific tasks.



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Chapter 5.1 Introduction to Applied NLP Tasks

Modular Pipelines vs End-t
End Models

- End-to-End: simpler, faster to prototype
- Modular pipelines: more explainable and controllable
- Enterprises often combine both approaches
- Design choice depends on scale and regulation

Monolithic vs. Modular LLM Systems

Characteristic	Monolithic LLM	Modular NLP Pipeline
Structure	Single large rectangular box	Multiple rectangular boxes connected left-to-right
Internal Detail	Minimal, just the label	NER, Classifier, Retriever, Generator
Flow	No arrows needed	NER → Classifier → Retriever → Generator
Comparison	Contrasts with multi-module pipeline	Contrasts with single-box system
Style	Academic / engineering blueprint	Academic / engineering blueprint
Palette	Monochrome (white and gray) on black	Monochrome (white and gray) on black
Accent Color	Optional (light blue or teal)	Optional (light blue or teal)
Typography	Sans-serif, neutral	Sans-serif, neutral
Design	Flat design	Flat design
Alignment	Horizontally aligned	Horizontally aligned
Intent	Visual comparison	Visual comparison

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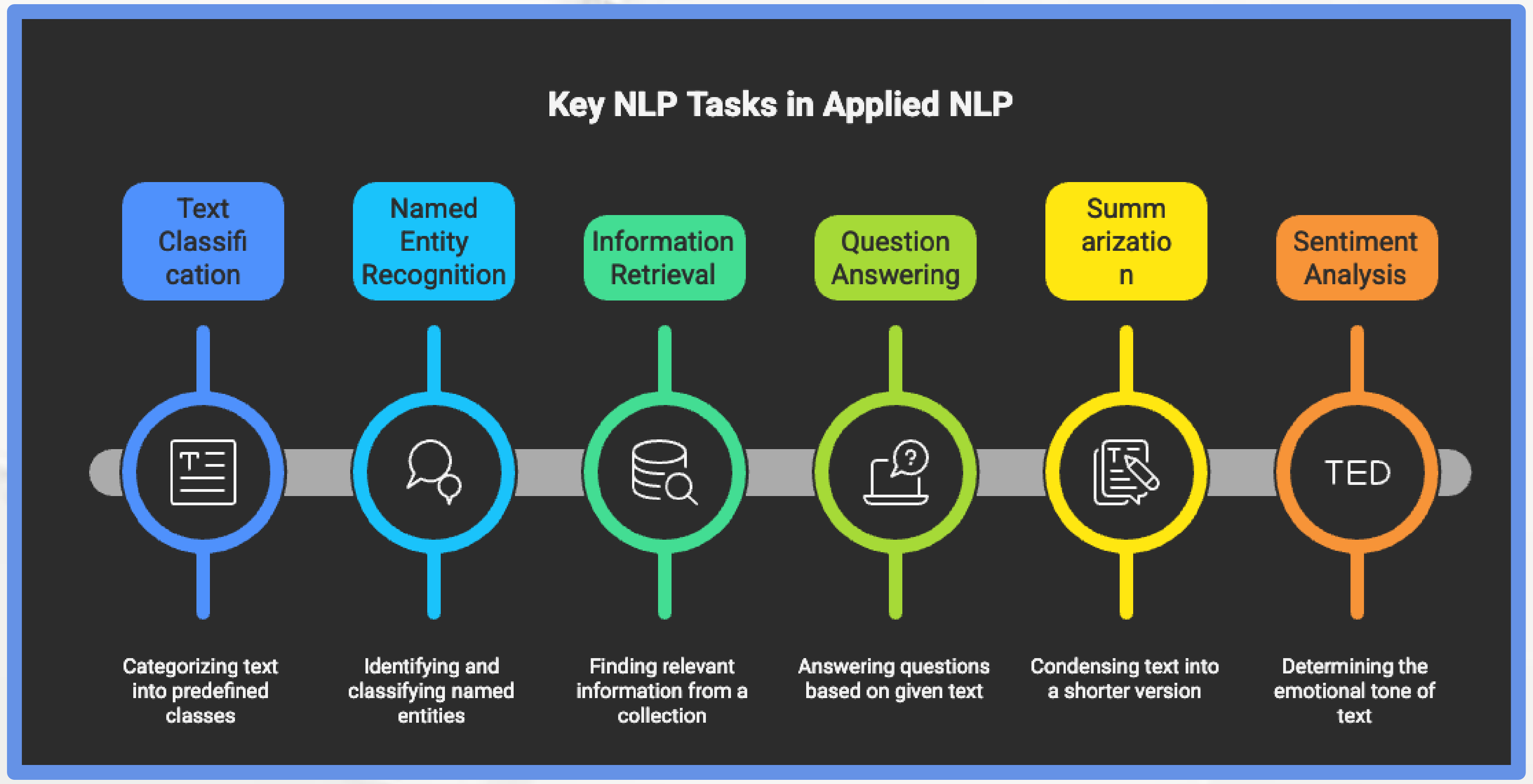
Chapter 5.1 Introduction to Applied NLP Tasks

Why This Module Matters

- Applied NLP turns language into action.
- Tasks define how AI creates value.
- Understanding task structure prevents misuse.
- Next chapters deep-dive into each task.



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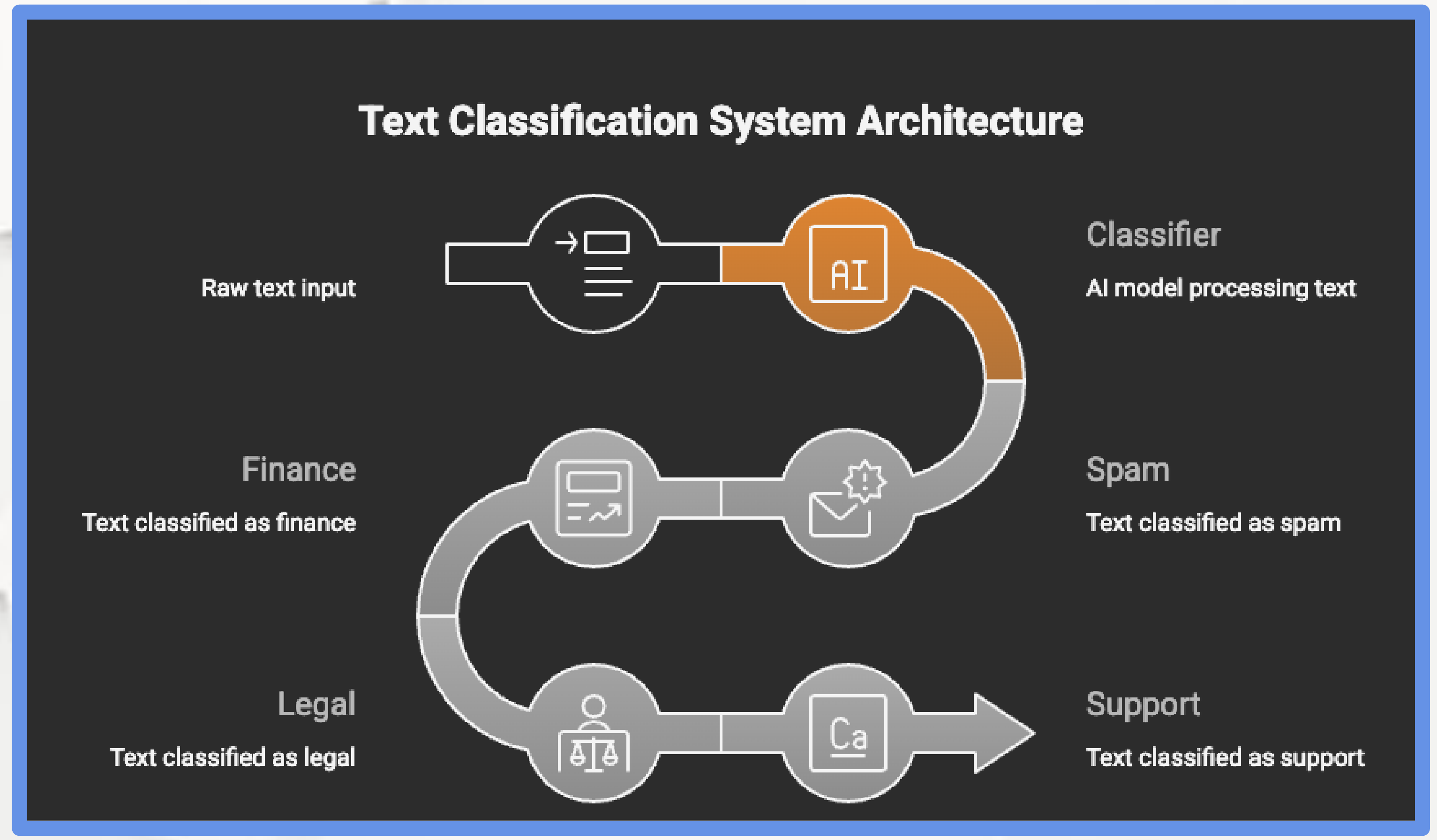
Chapter 5.2 Text Classification

What is Text Classification?

- Task: assign predefined labels to text.
- One of the oldest and most deployed NLP tasks.
- Foundation for moderation, routing, filtering, analytics.
- Used in almost every production NLP system.



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Chapter 5.2 Text Classification

Classification Problem Types

- Binary: two exclusive classes (Yes / No)
- Multi-class: one label from many
- Multi-label: multiple labels per document
- Correct framing defines model architecture

Classification Paradigms

Characteristic	Output Boxes	Active Outputs	Arrow	Title
Binary Classification	Two	One	Single	Binary Classification
Multi-class Classification	Multiple	One	To set	Multi-class Classification
Multi-label Classification	Multiple	Several	To set	Multi-label Classification

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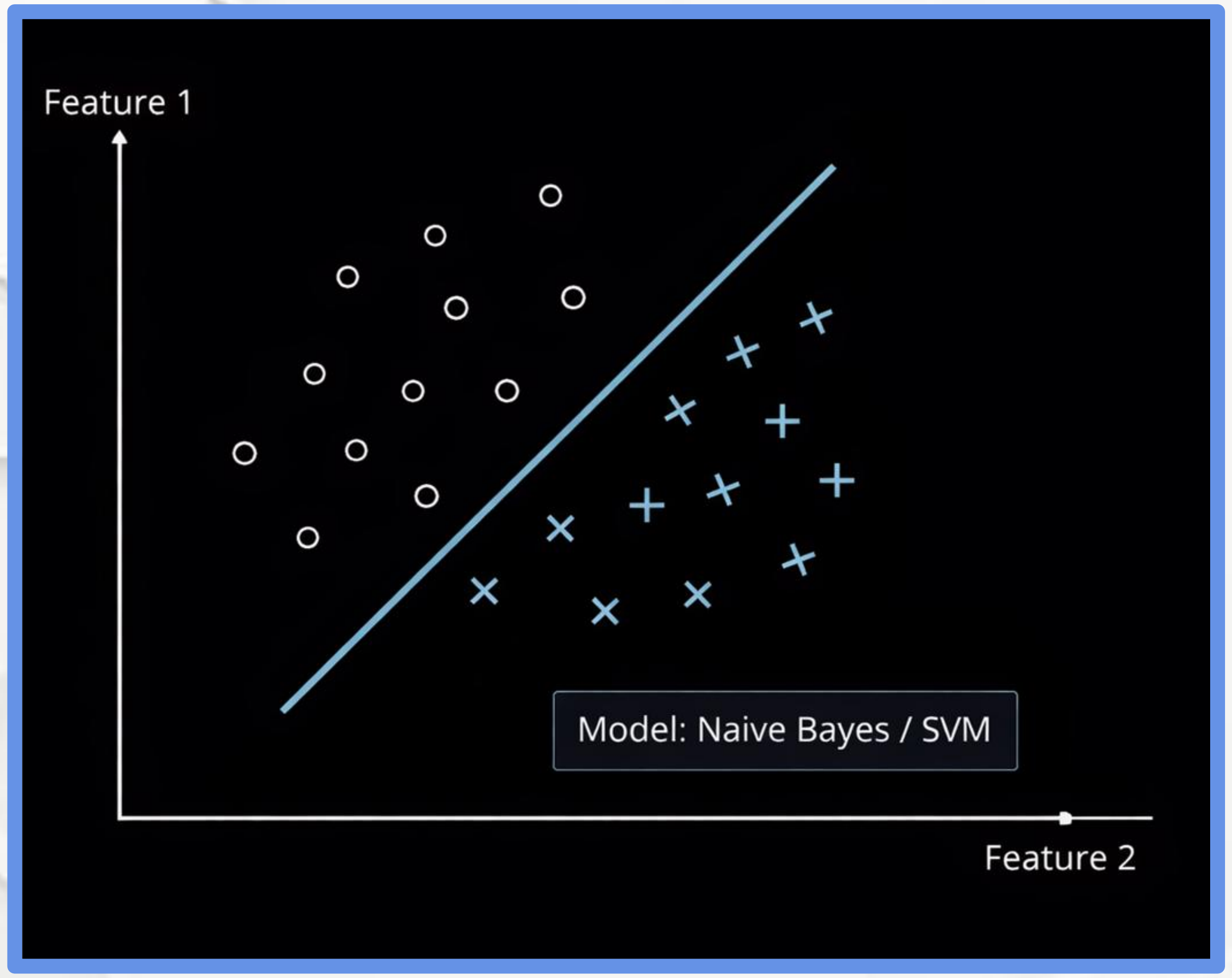
Chapter 5.2 Text Classification

Traditional Classification Methods

- Based on word frequency statistics.
- Require manual feature engineering.
- Still useful for small or structured datasets.
- Serve as strong baselines.



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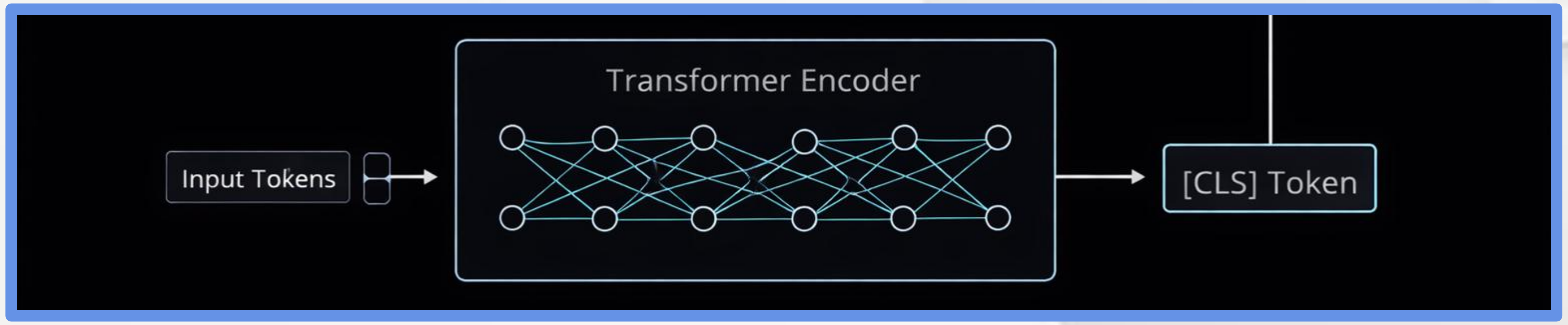
Chapter 5.2 Text Classification

Deep Learning Classification

- Models learn features automatically.
- Capture semantic meaning, not just keywords.
- Transformers are the modern standard.
- Context-aware classification.



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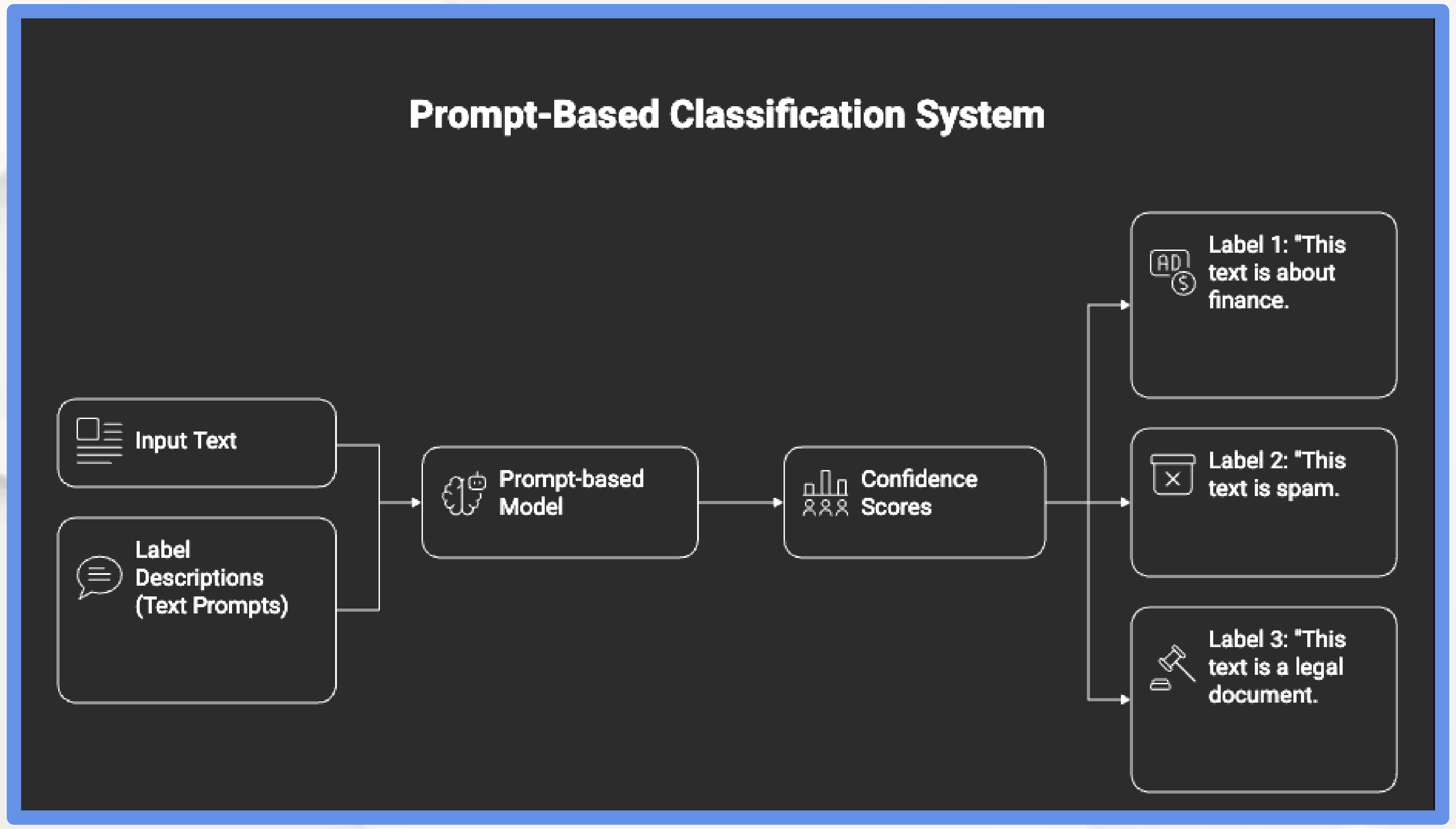
Chapter 5.2 Text Classification

Zero-Shot Classification

- No task-specific training required.
- Uses natural language label descriptions.
- Extremely fast to deploy.
- Ideal for dynamic taxonomies.



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Chapter 5.2 Text Classification

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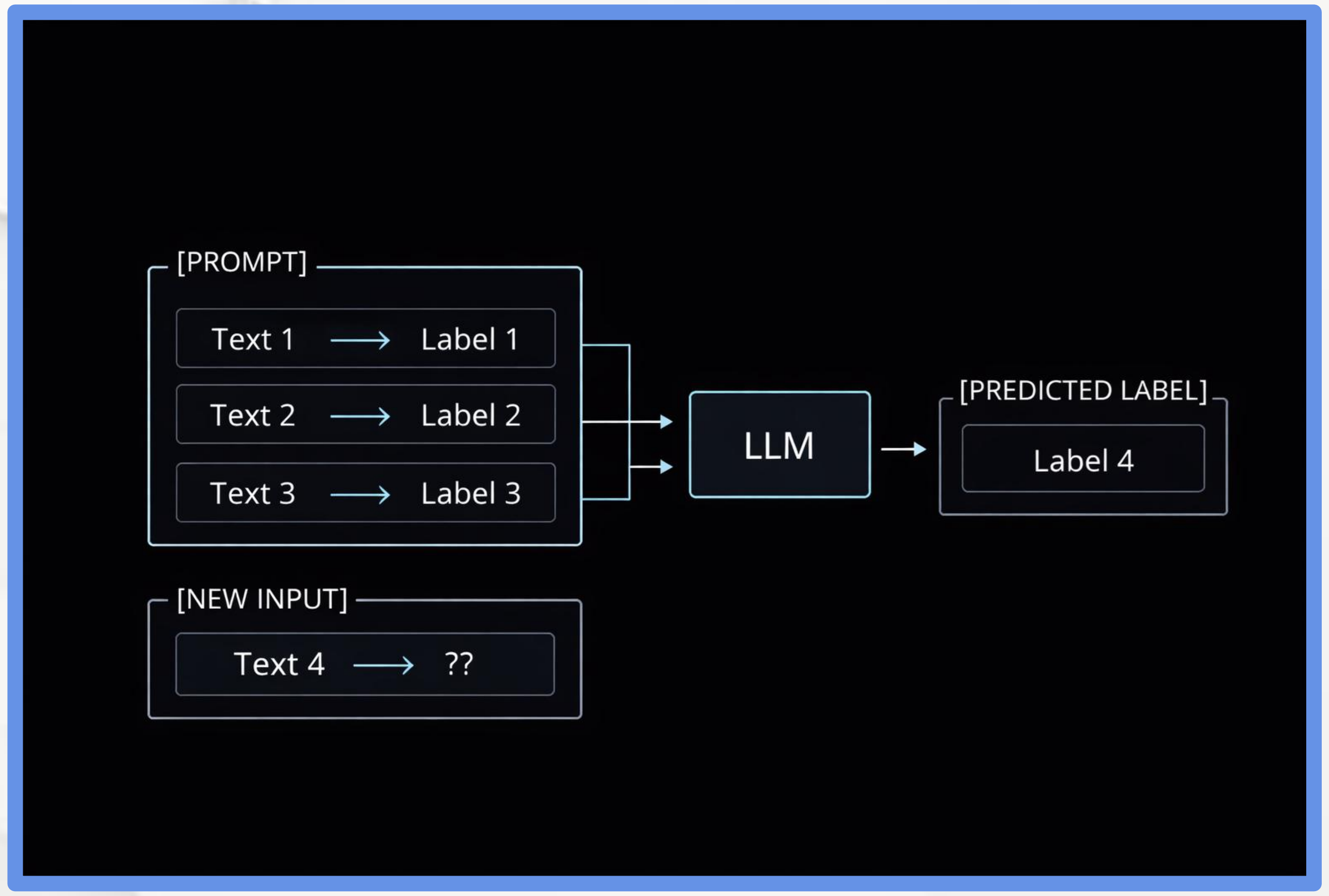

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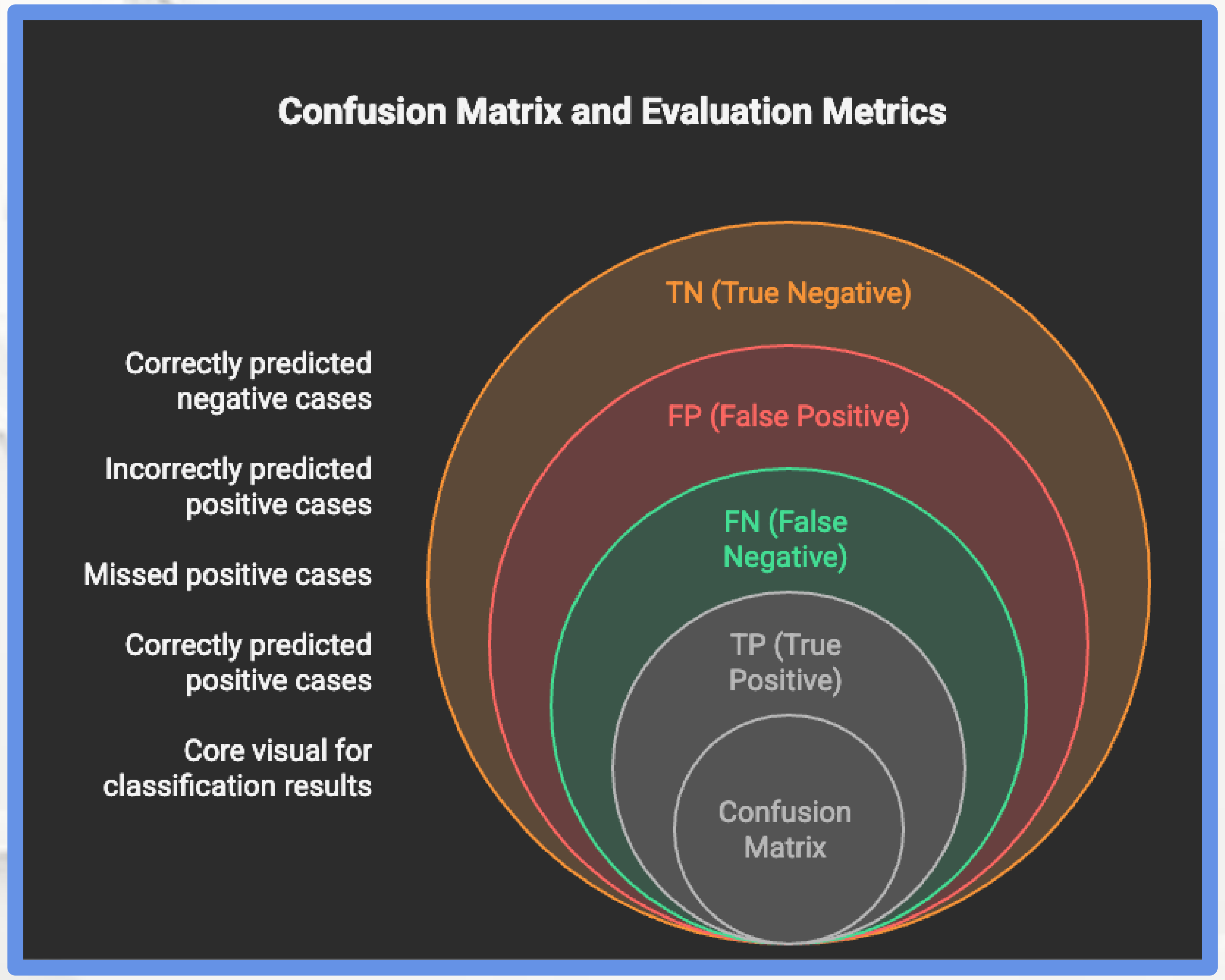
Chapter 5.2 Text Classification

Evaluation Metrics

- Accuracy is often misleading
- Precision controls false positives
- Recall controls false negatives
- F1-score balances both
- Metric choice depends on risk



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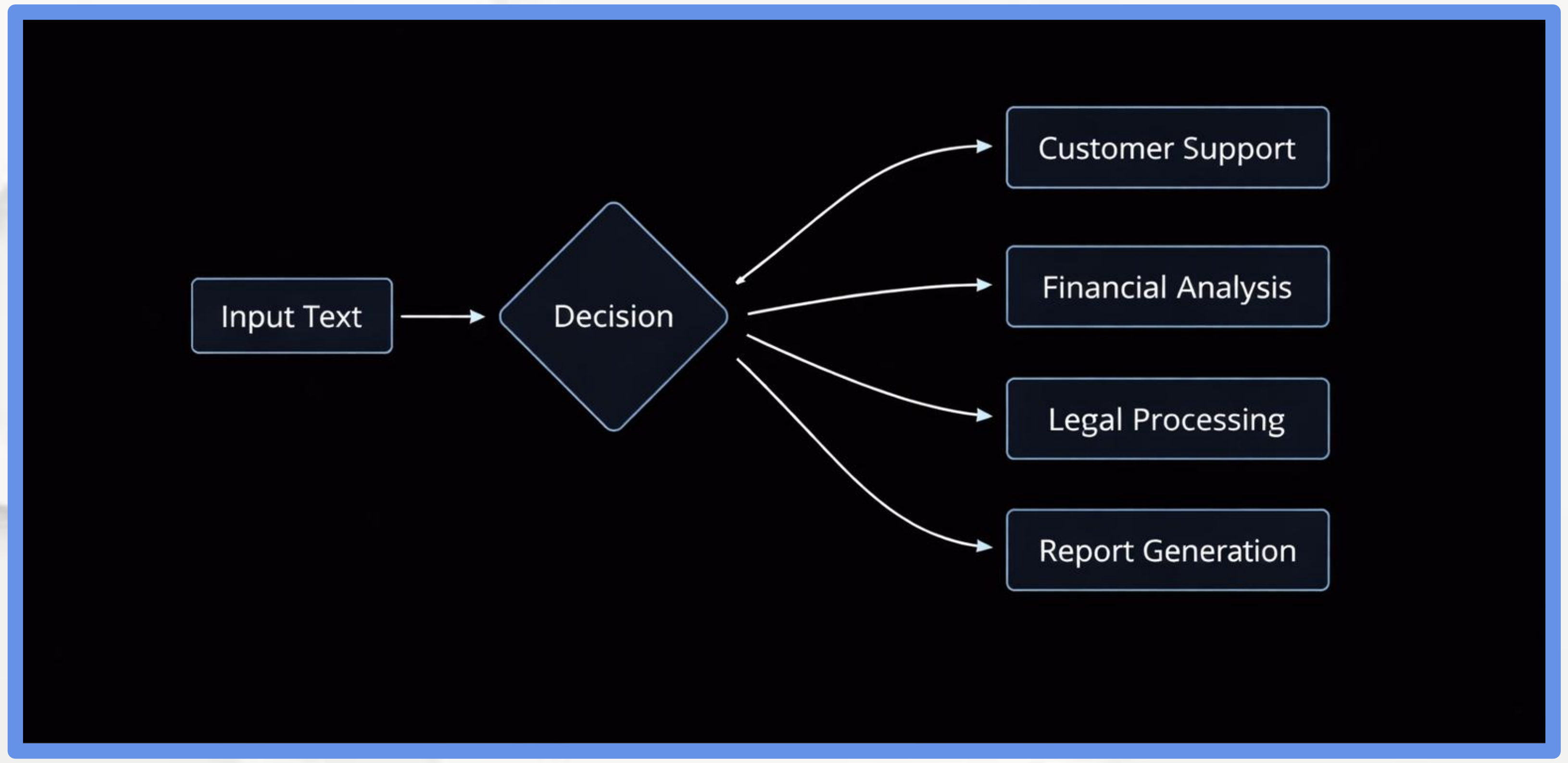
Chapter 5.2 Text Classification

Key Takeaways

- Classification transforms language into decisions
- Architecture choice matters more than model size
- Zero-shot enables rapid deployment
- Metrics define real-world success



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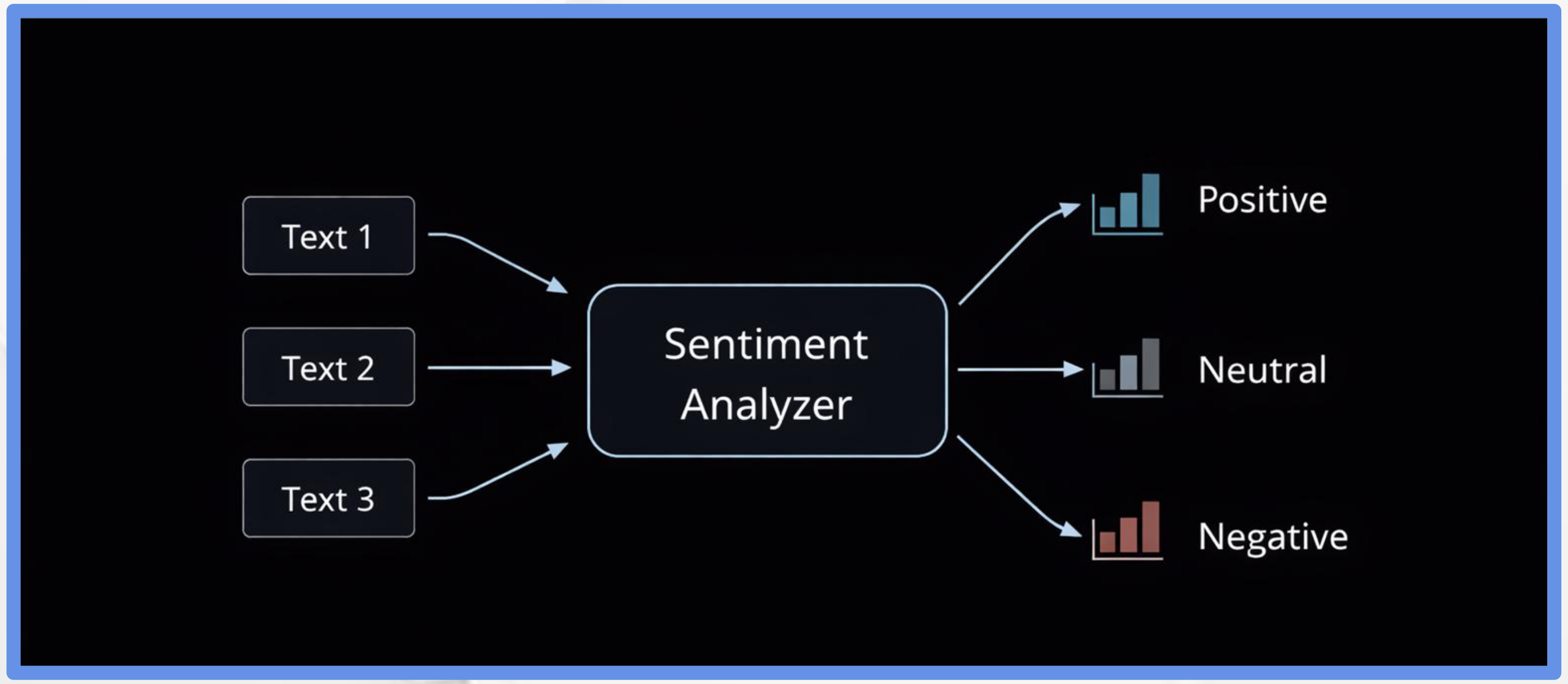
Chapter 5.3 Understanding User Intent: Sentiment Analysis

What is Sentiment Analysis?

- Task: detect subjective intent in text
- Focuses on opinions, emotions, attitudes
- Specialized form of text classification
- Core signal for user and market intelligence



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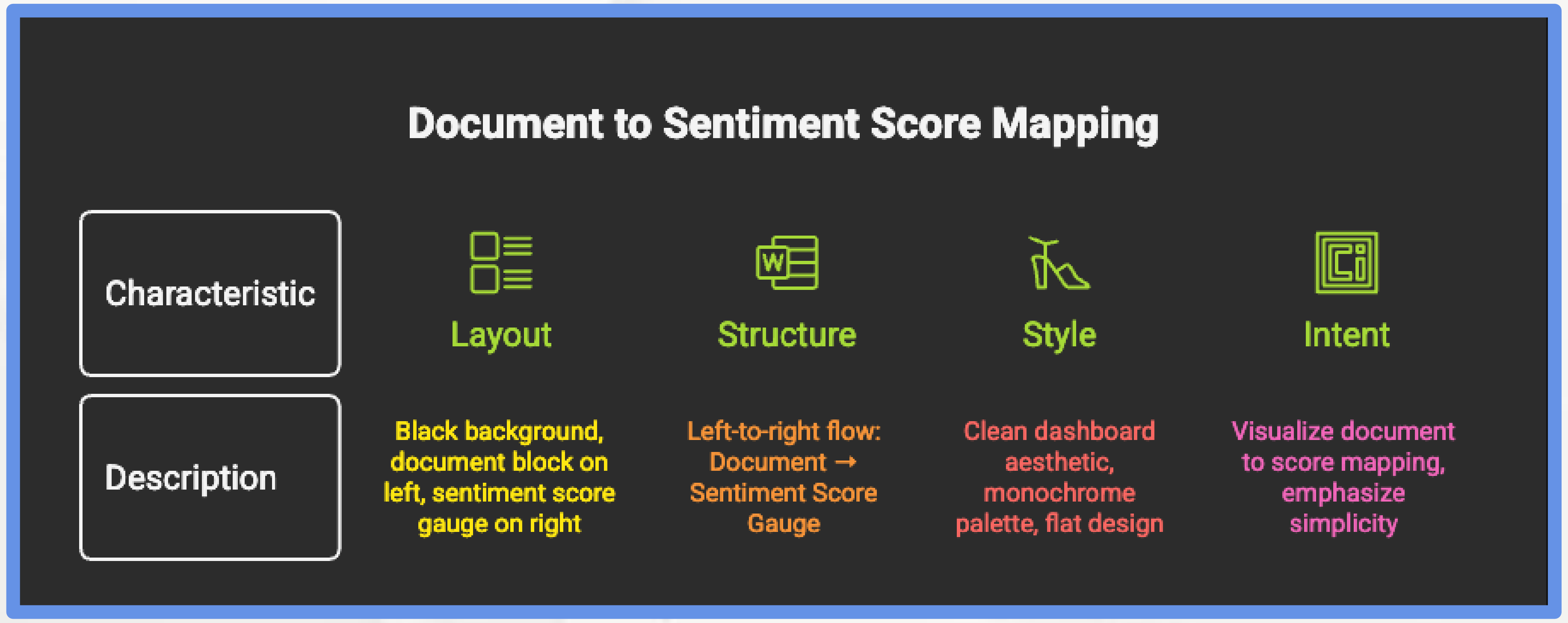
Chapter 5.3 Understanding User Intent: Sentiment Analysis

Document-Level Sentiment

- One sentiment for the entire document
- Simple and computationally efficient
- Works well for short reviews or posts
- Loses internal nuance



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Chapter 5.3 Understanding User Intent: Sentiment Analysis

Document-Level Sentiment

- One sentiment for the entire document
- Simple and computationally efficient
- Works well for short reviews or posts
- Loses internal nuance

Document to Sentiment Score Mapping

Characteristic	Layout	Structure	Style	Intent
Description	Black background, document block on left, sentiment score gauge on right	Left-to-right flow: Document → Sentiment Score Gauge	Clean dashboard aesthetic, monochrome palette, flat design	Visualize document to score mapping, emphasize simplicity

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Chapter 5.3 Understanding User Intent: Sentiment Analysis

Aspect-Based Sentiment Analysis (ABSA)

- Sentiment linked to specific entities or features
- Enables fine-grained insight
- Essential for product and brand analysis
- Industry standard for actionable feedback

Aspect-Based Sentiment Analysis

Characteristic	Description
Layout	Sentence in center, aspect nodes around
Polarity Outputs	Arrow from aspect to sentiment label
Structure	Sentence is central reference, aspects branch
Color-Coded Arrows	Blue for positive, gray for neutral, red for negative
Style	Technical NLP annotation, flat design
Intent	Visualize aspect-based sentiment analysis

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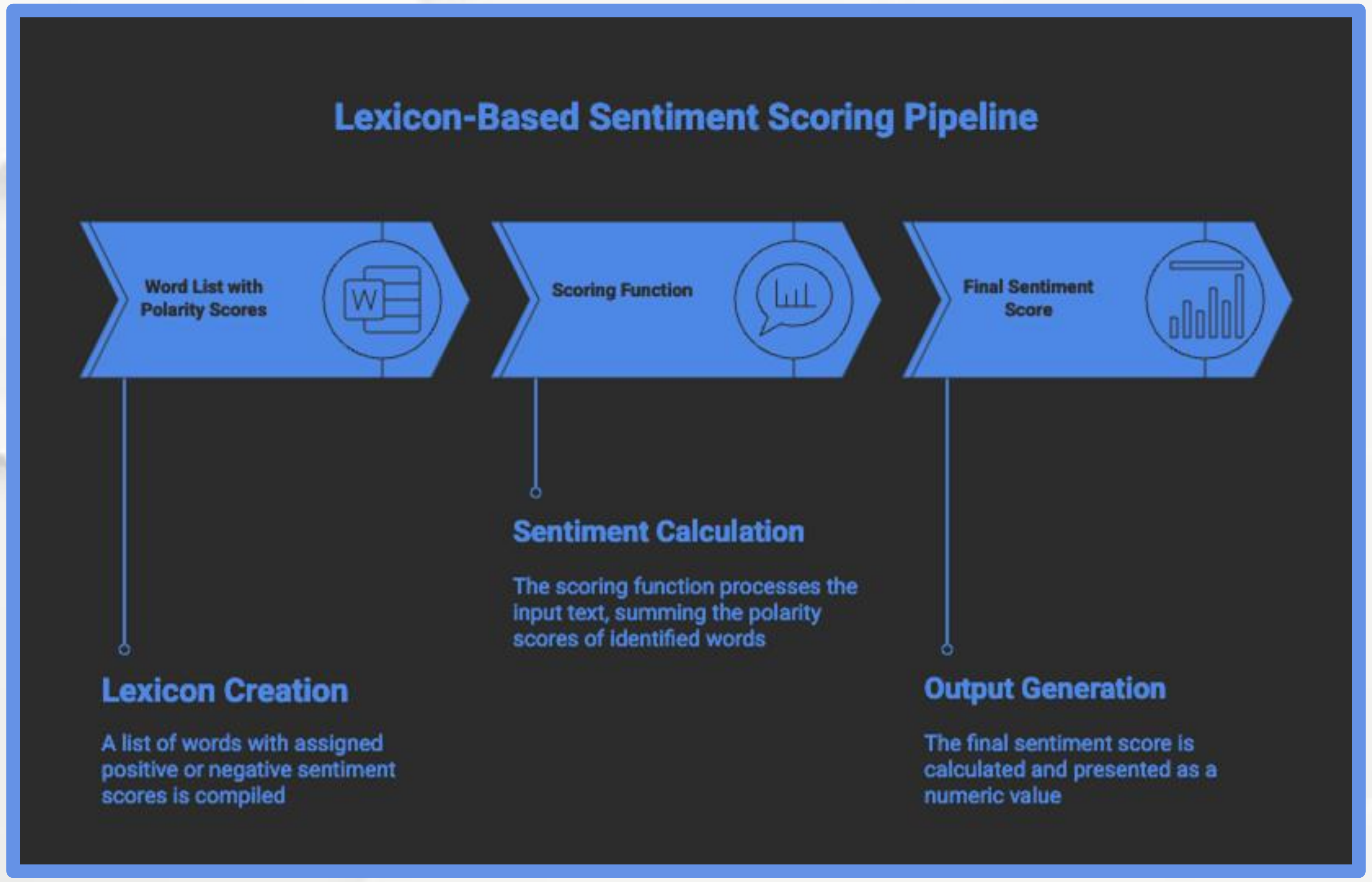
Chapter 5.3 Understanding User Intent: Sentiment Analysis

Lexicon-Based Approaches

- Use predefined polarity dictionaries
- Fully explainable
- Fast and lightweight
- Fail under context, irony, or negation



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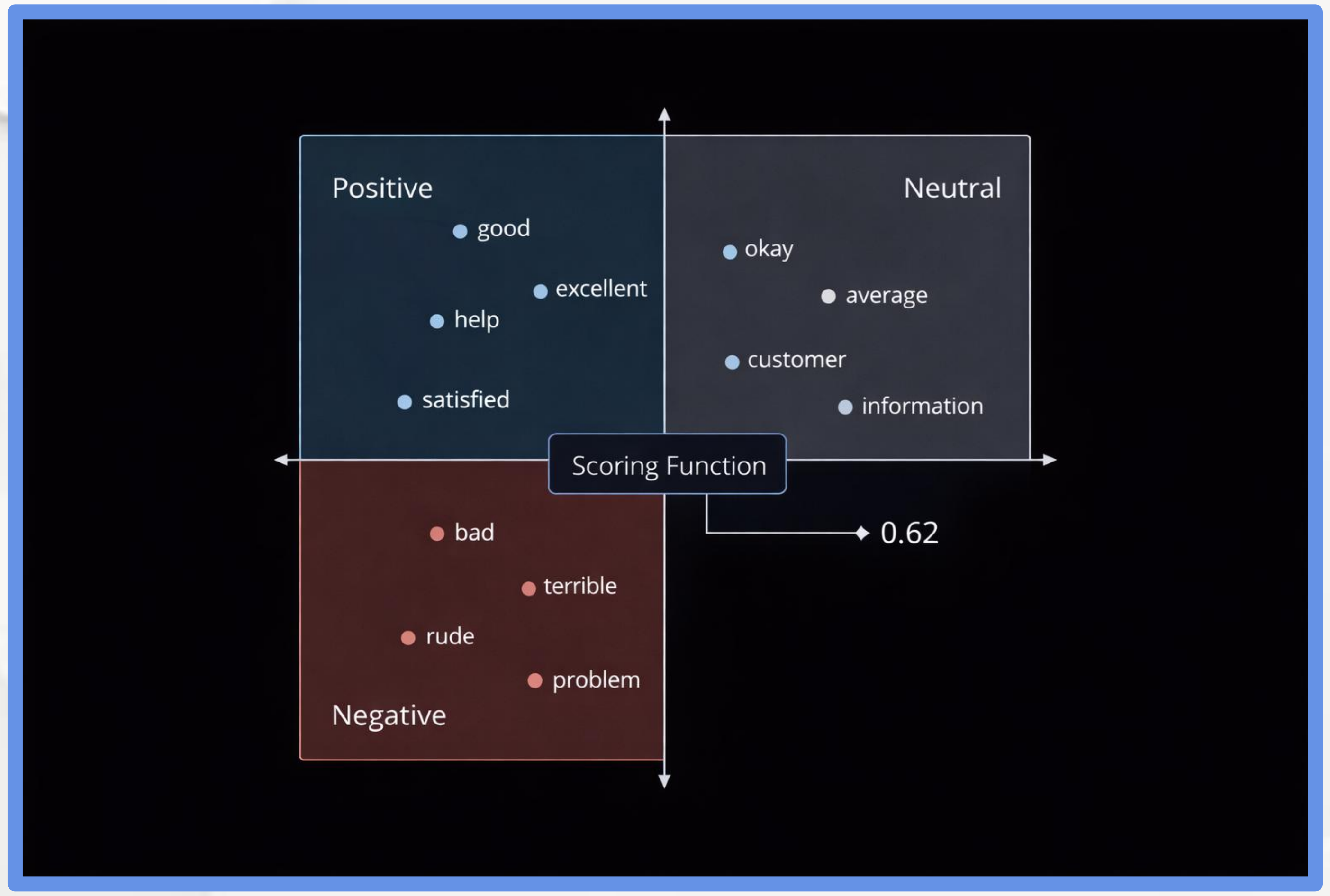
Chapter 5.3 Understanding User Intent: Sentiment Analysis

Embedding-Based Sentiment Models

- Meaning represented in latent vector space
- Generalizes beyond exact keywords
- Robust to synonyms and paraphrases
- Foundation for transformer-based sentiment



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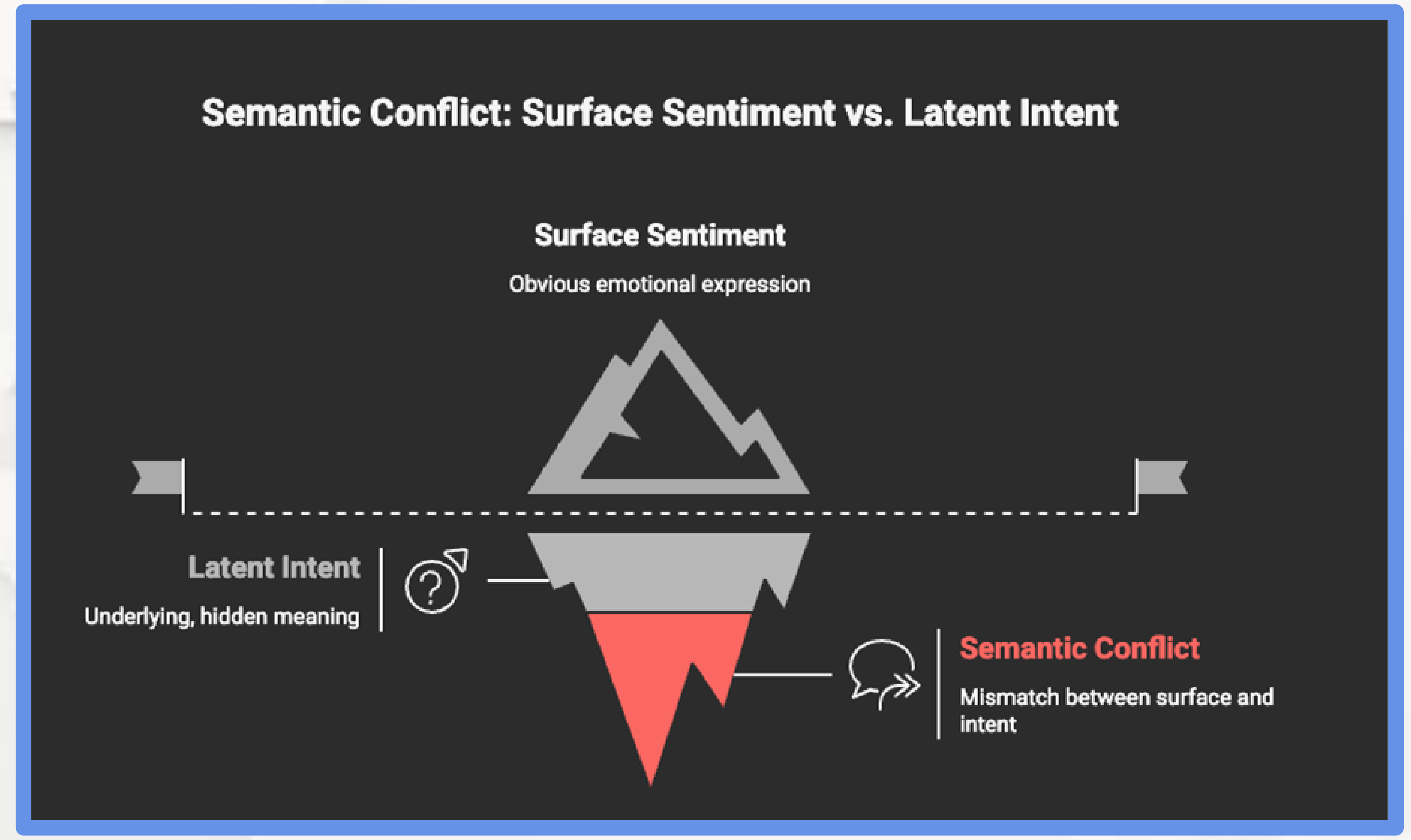
Chapter 5.3 Understanding User Intent: Sentiment Analysis

Negation and Sarcasm Challenges

- Negation flips polarity meaning
- Sarcasm contradicts surface wording
- Requires long-range context
- Transformers outperform classical models



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Chapter 5.3 Understanding User Intent: Sentiment Analysis

Sentiment in Market Intelligence

- Meaning represented in latent vector space
- Generalizes beyond exact keywords
- Robust to synonyms and paraphrases
- Foundation for transformer-based sentiment

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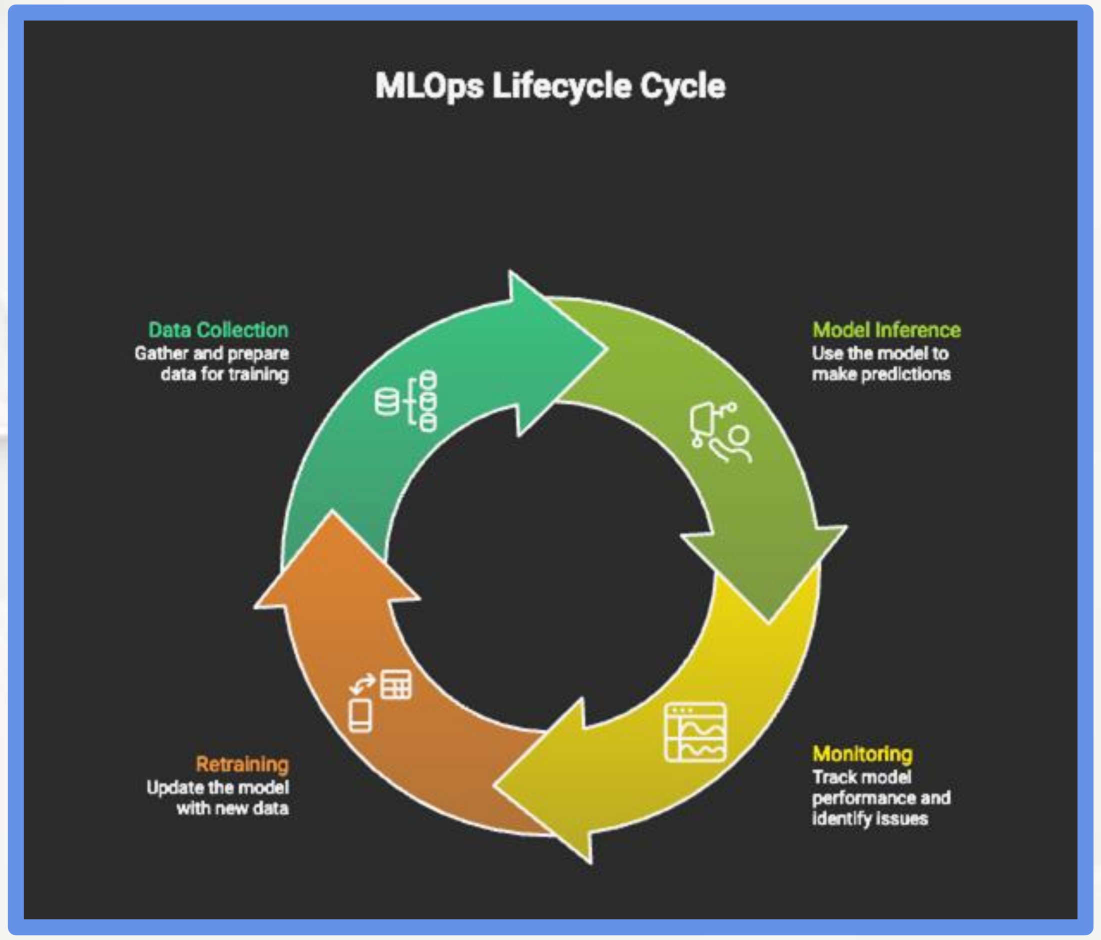
Chapter 5.3 Understanding User Intent: Sentiment Analysis

Operational Considerations

- Domain-specific language matters
- Models degrade with cultural drift
- Requires periodic re-evaluation
- Sentiment ≠ truth



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Chapter 5.3 Understanding User Intent: Sentiment Analysis

Key Takeaways

- Sentiment reveals user intent.
- Aspect-level analysis enables action.
- Context is the main technical challenge.
- Critical signal for applied NLP systems.

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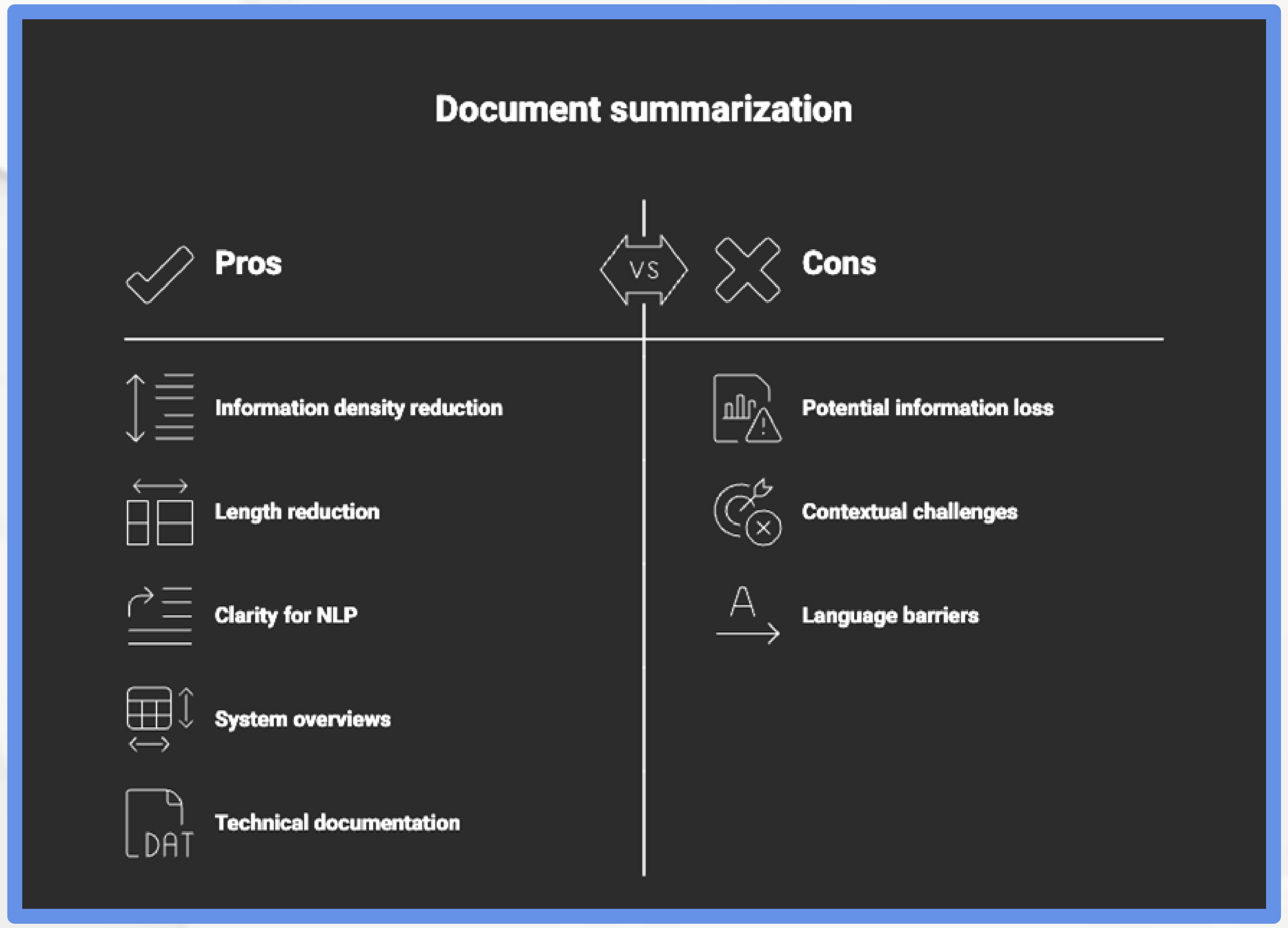
Chapter 5.4 Reducing Information Overload: Text Summarization

What is Text Summarization?

- Task: compress long text into key information.
- Preserves meaning while reducing length.
- Essential for scale and decision speed.
- Widely used in legal, news, intelligence.



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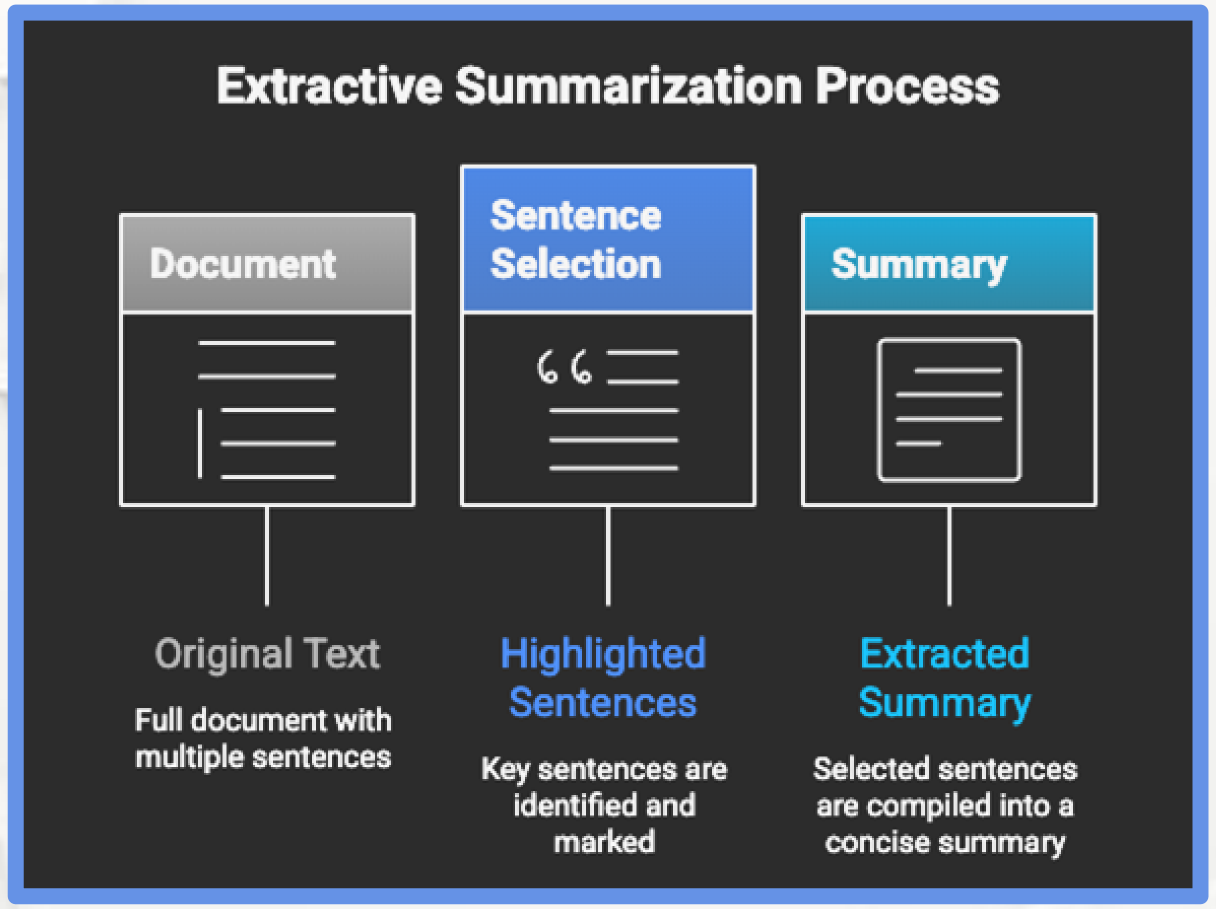
Chapter 5.4 Reducing Information Overload: Text Summarization

Extractive Summarization

- Task: compress long text into key information
- Preserves meaning while reducing length
- Essential for scale and decision speed
- Widely used in legal, news, intelligence



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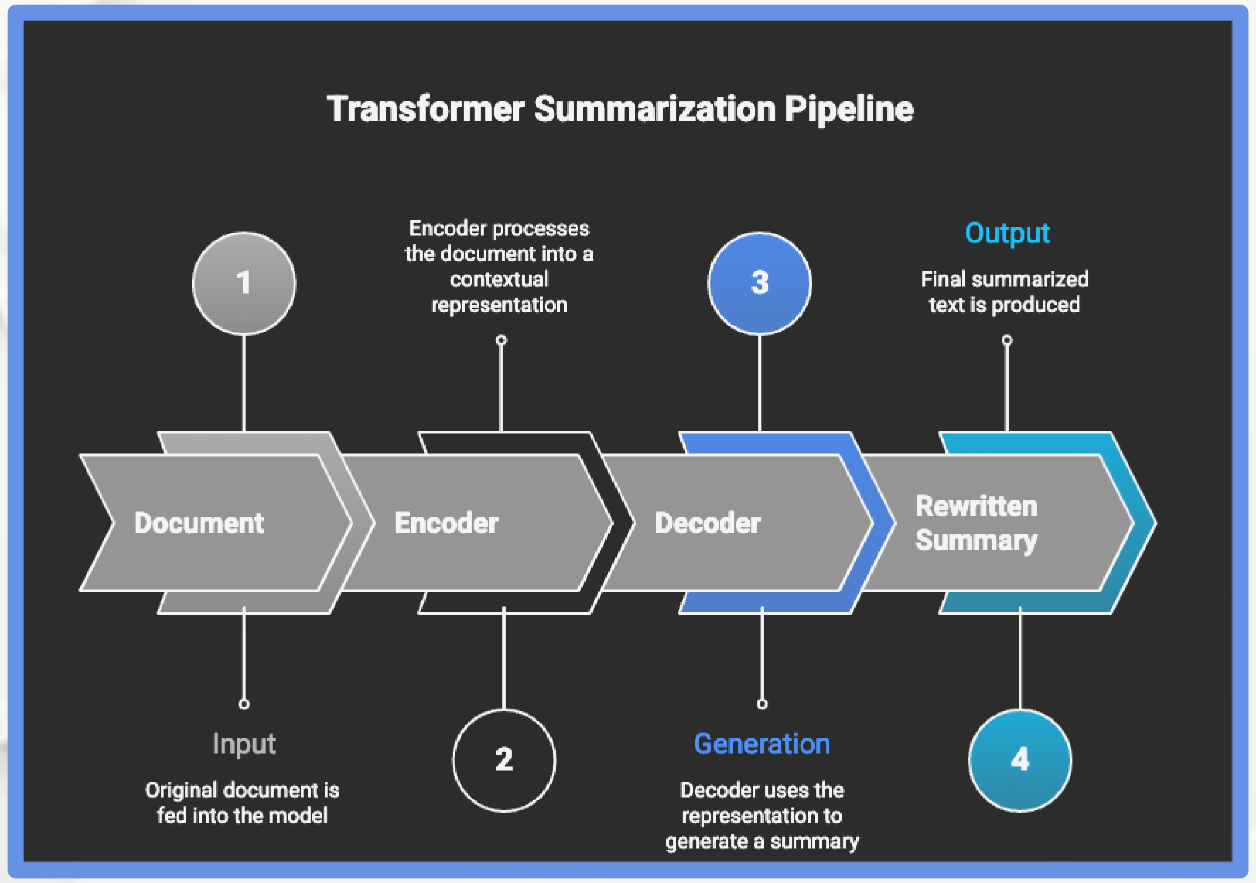
Chapter 5.4 Reducing Information Overload: Text Summarization

Abstractive Summarization

- Generates new sentences
- Higher information density
- More natural language
- Risk of hallucination



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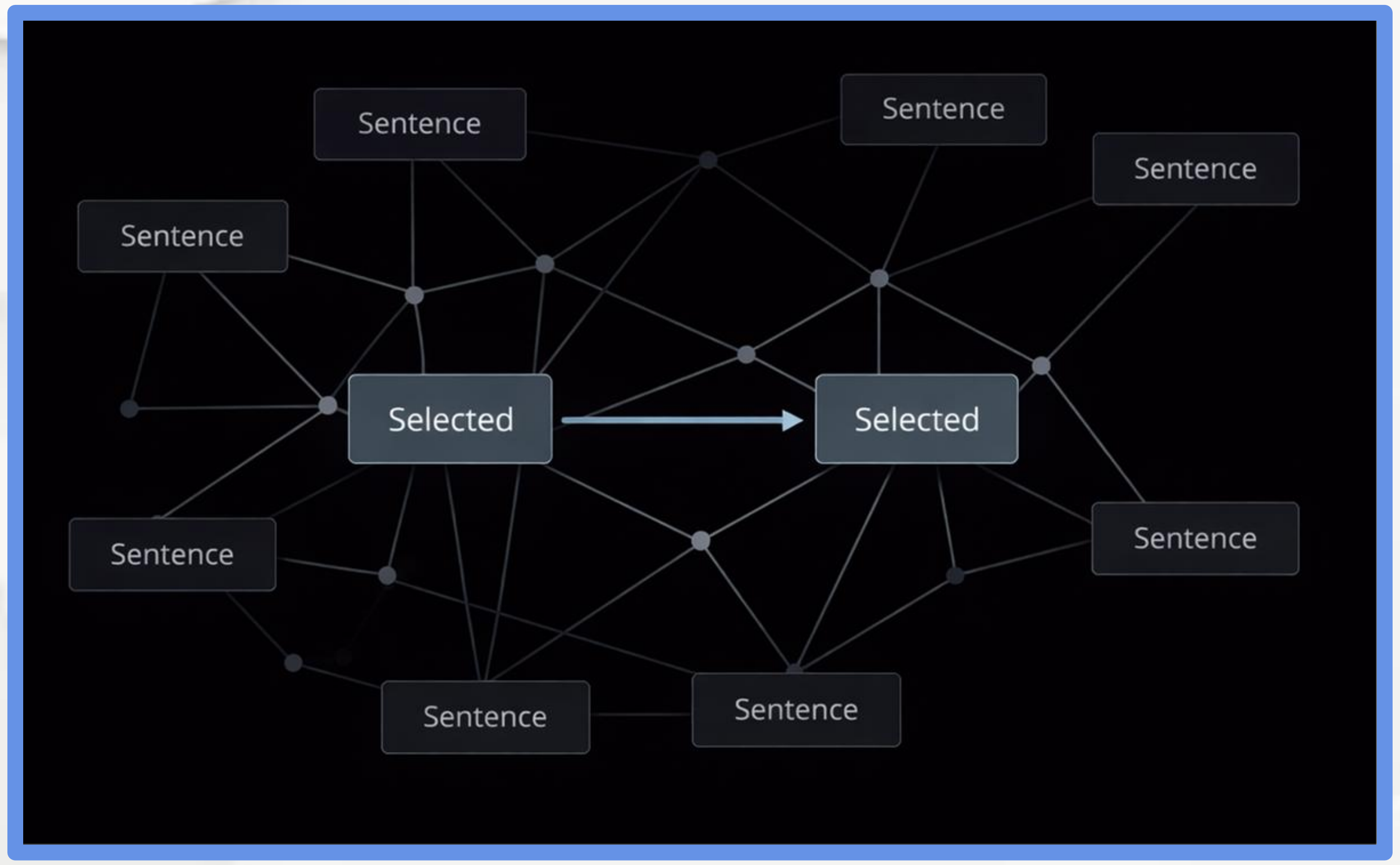
Chapter 5.4 Reducing Information Overload: Text Summarization

Traditional Extractive Techniques

- Graph-based ranking (TextRank)
- Sentence similarity scoring
- No deep semantics
- Still useful for formal text



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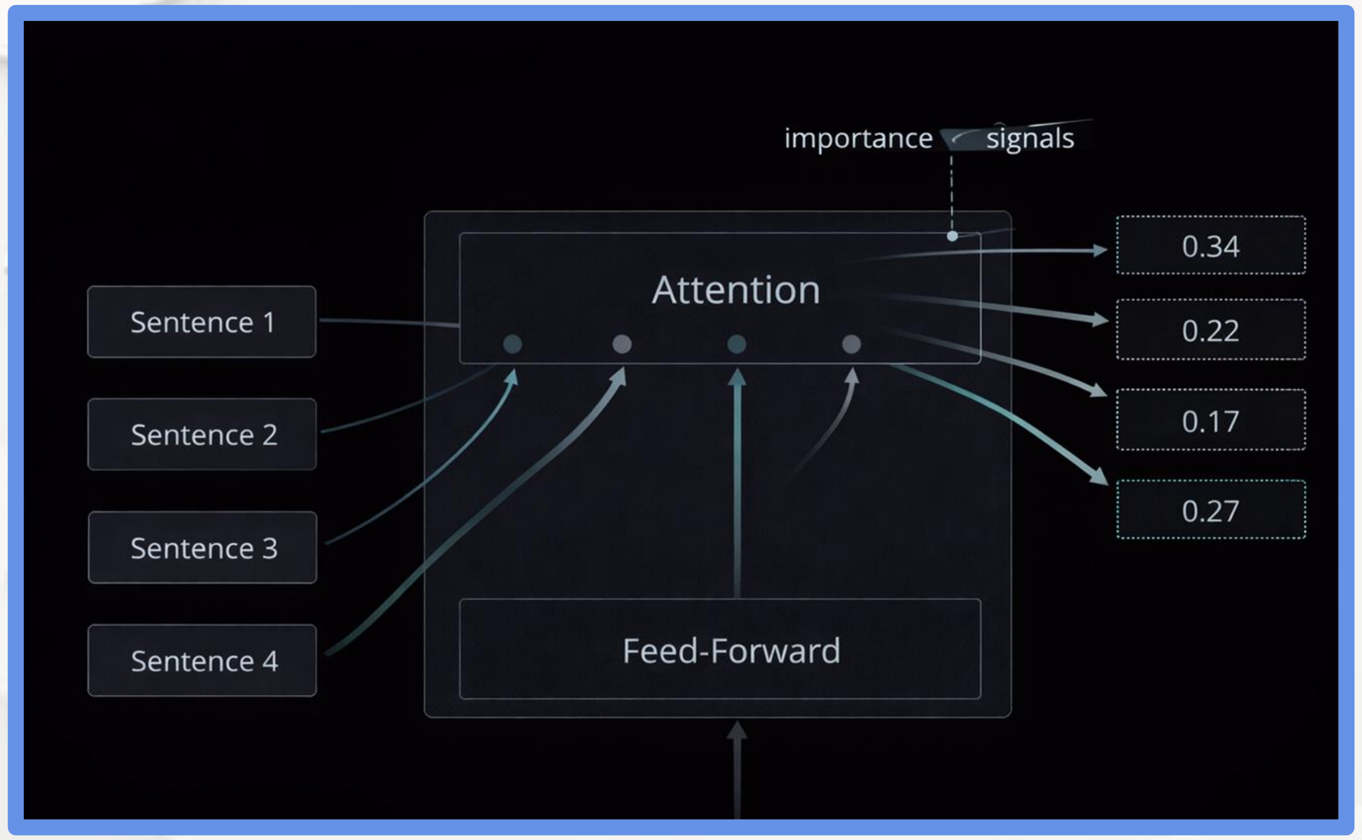
Chapter 5.4 Reducing Information Overload: Text Summarization

Transformer-Based Summarization

- Uses full-document context
- Learns semantic importance
- State-of-the-art performance
- Dominant in production systems



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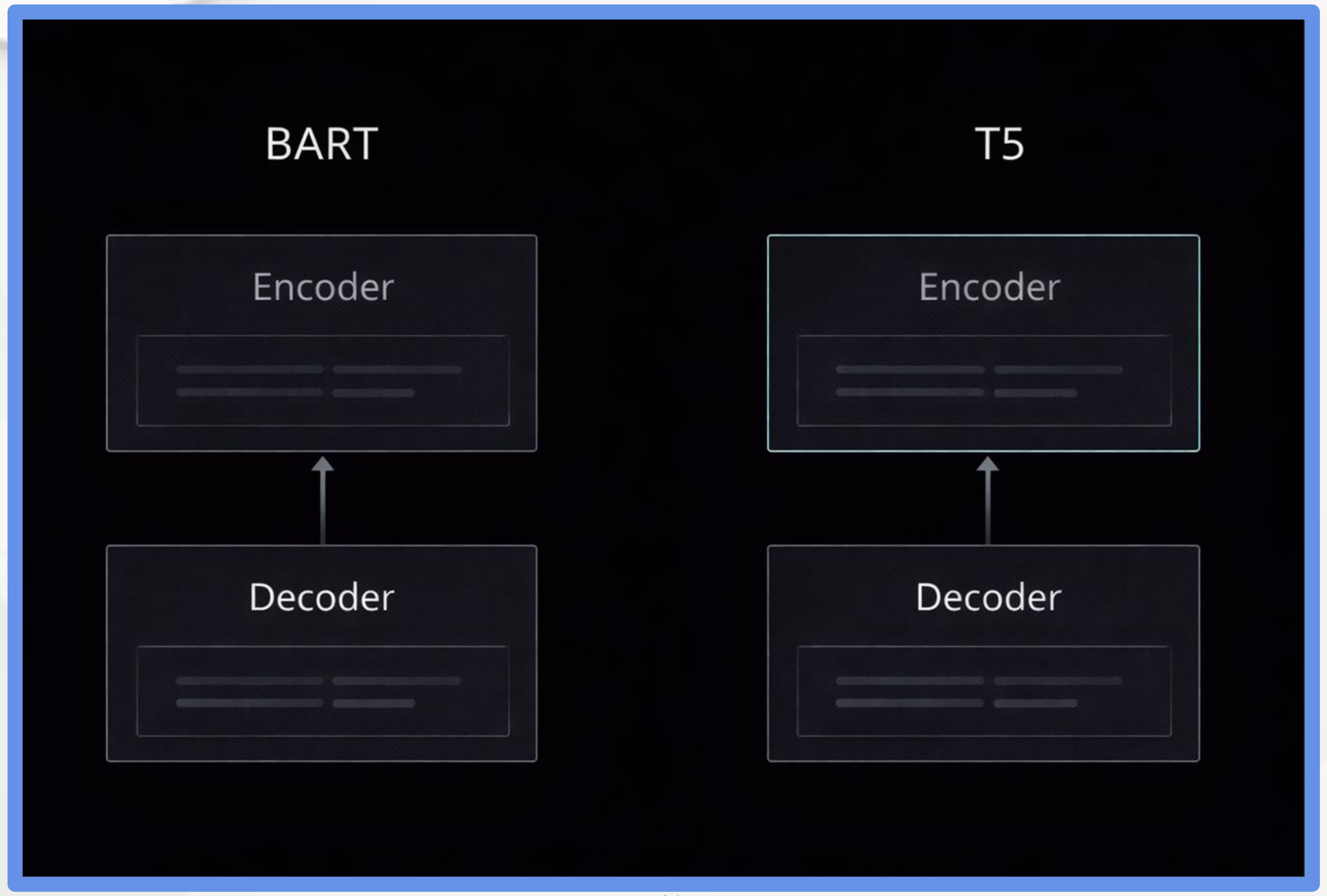
Chapter 5.4 Reducing Information Overload: Text Summarization

BART and T5 Models

- UEncoder—Decoder architectures
- Pre-trained for text reconstruction
- Strong performance on long documents
- Industry standard for summarization



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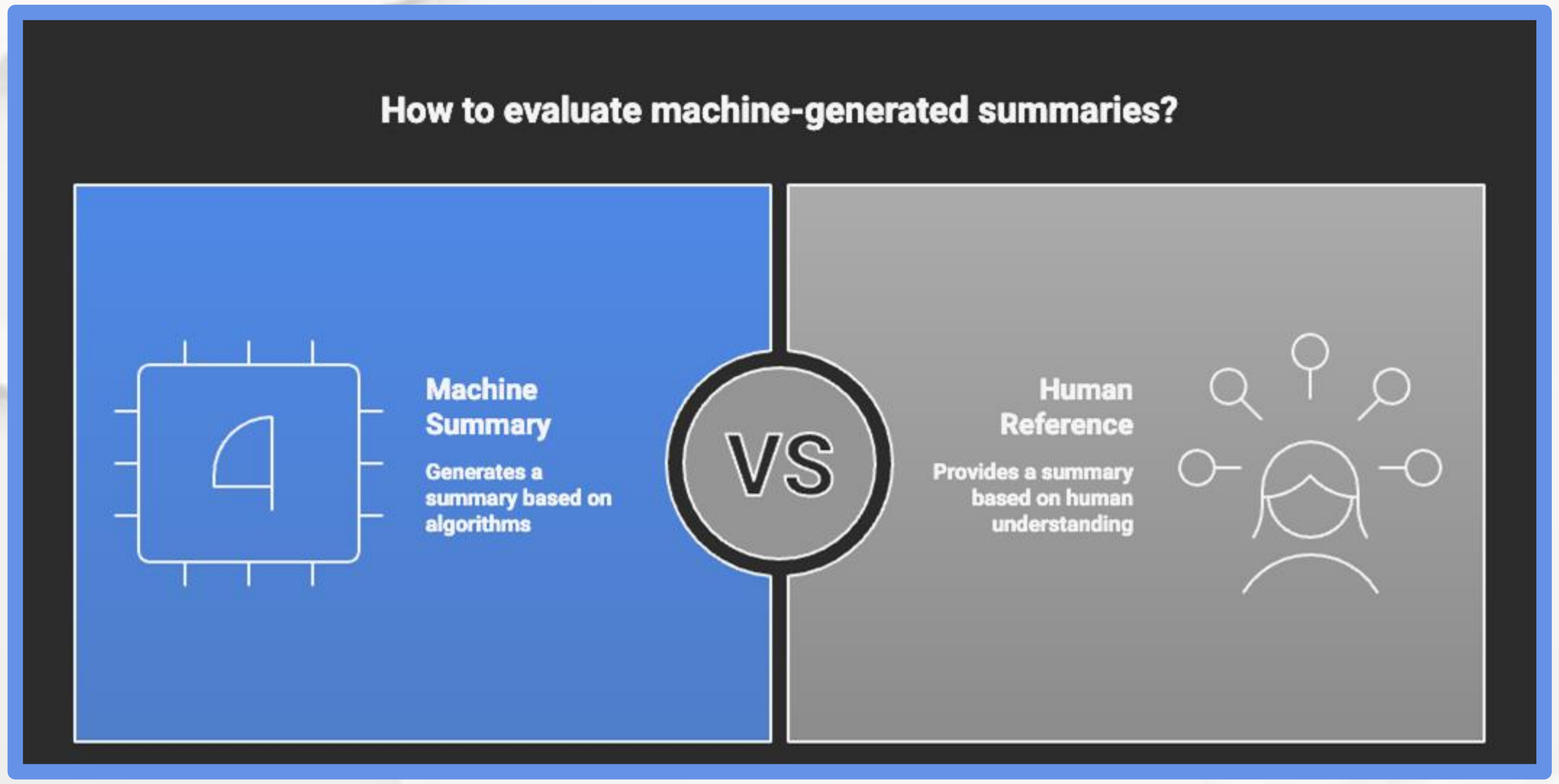
Chapter 5.4 Reducing Information Overload: Text Summarization

Evaluation Metrics (ROUGE)

- Measures overlap with human summaries
- ROUGE-1, ROUGE-2, ROUGE-L
- Focuses on recall
- Does not ensure factuality



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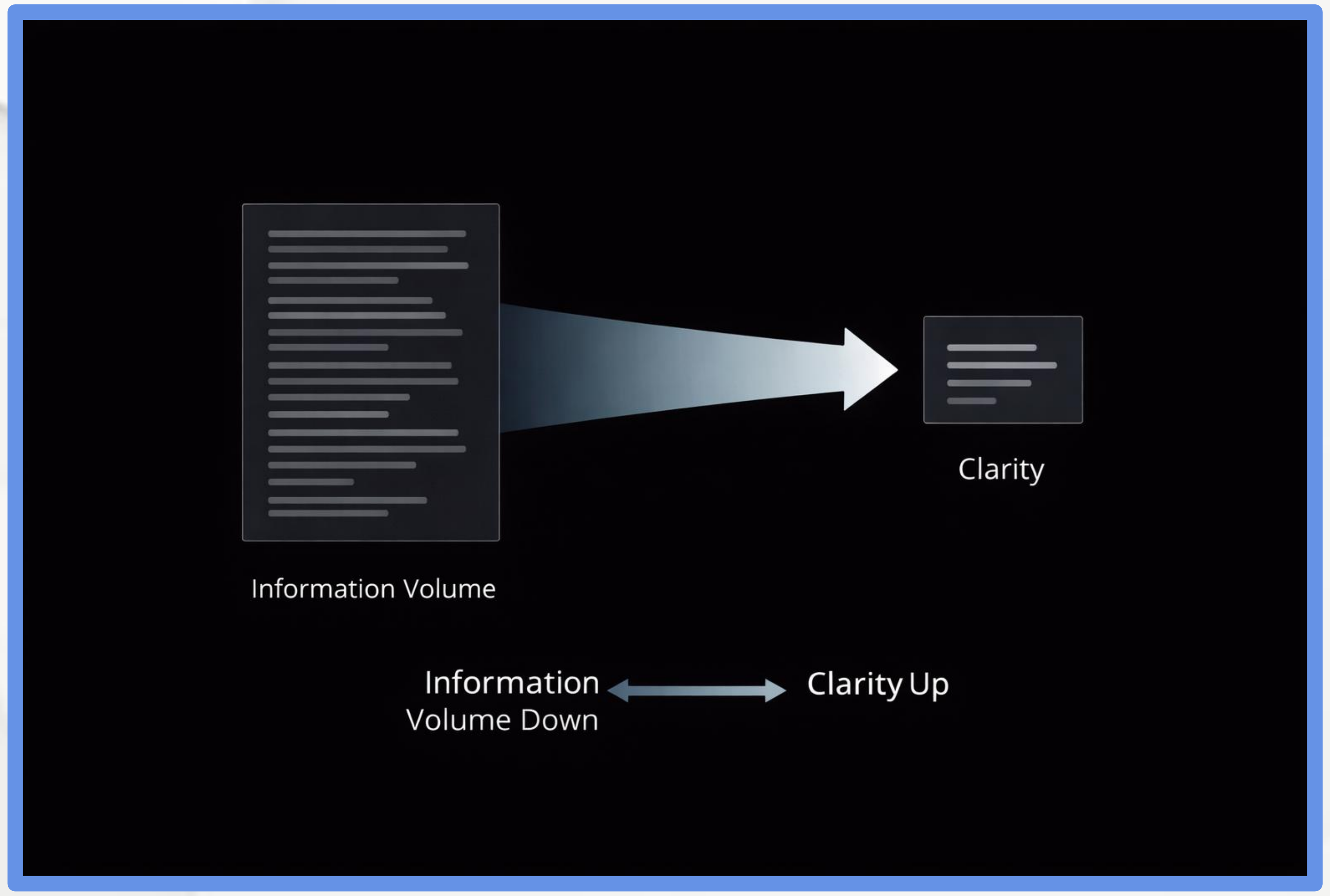
Chapter 5.4 Reducing Information Overload: Text Summarization

Key Takeaways

- Summarization manages information overload
- Extractive = safer, abstractive = denser
- Transformers dominate modern systems



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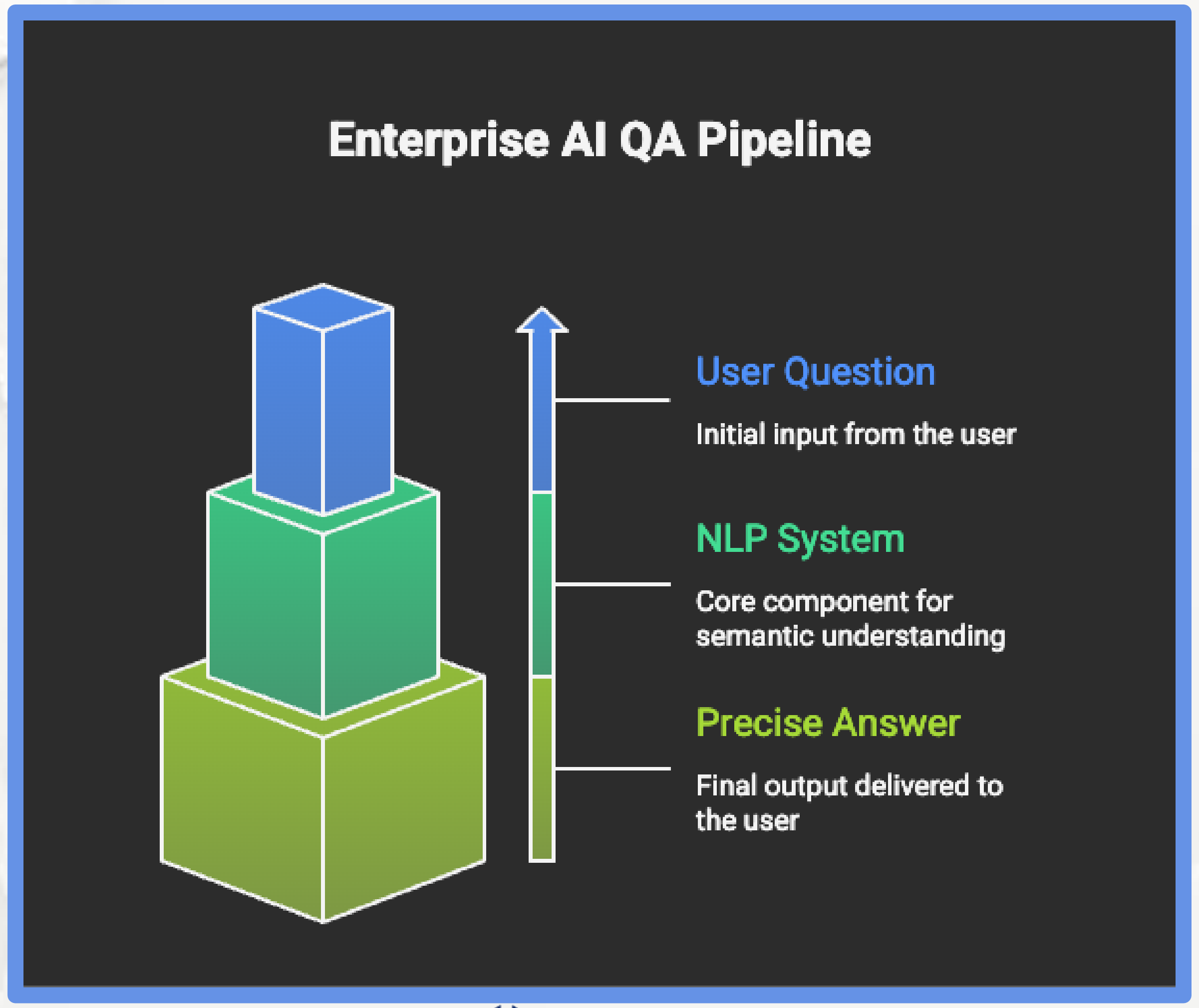
Chapter 5.5 Information Retrieval: Question Answering Systems

What is Question Answering (QA)?

- Goes beyond keyword search
- Requires language understanding
- Core of modern AI assistants



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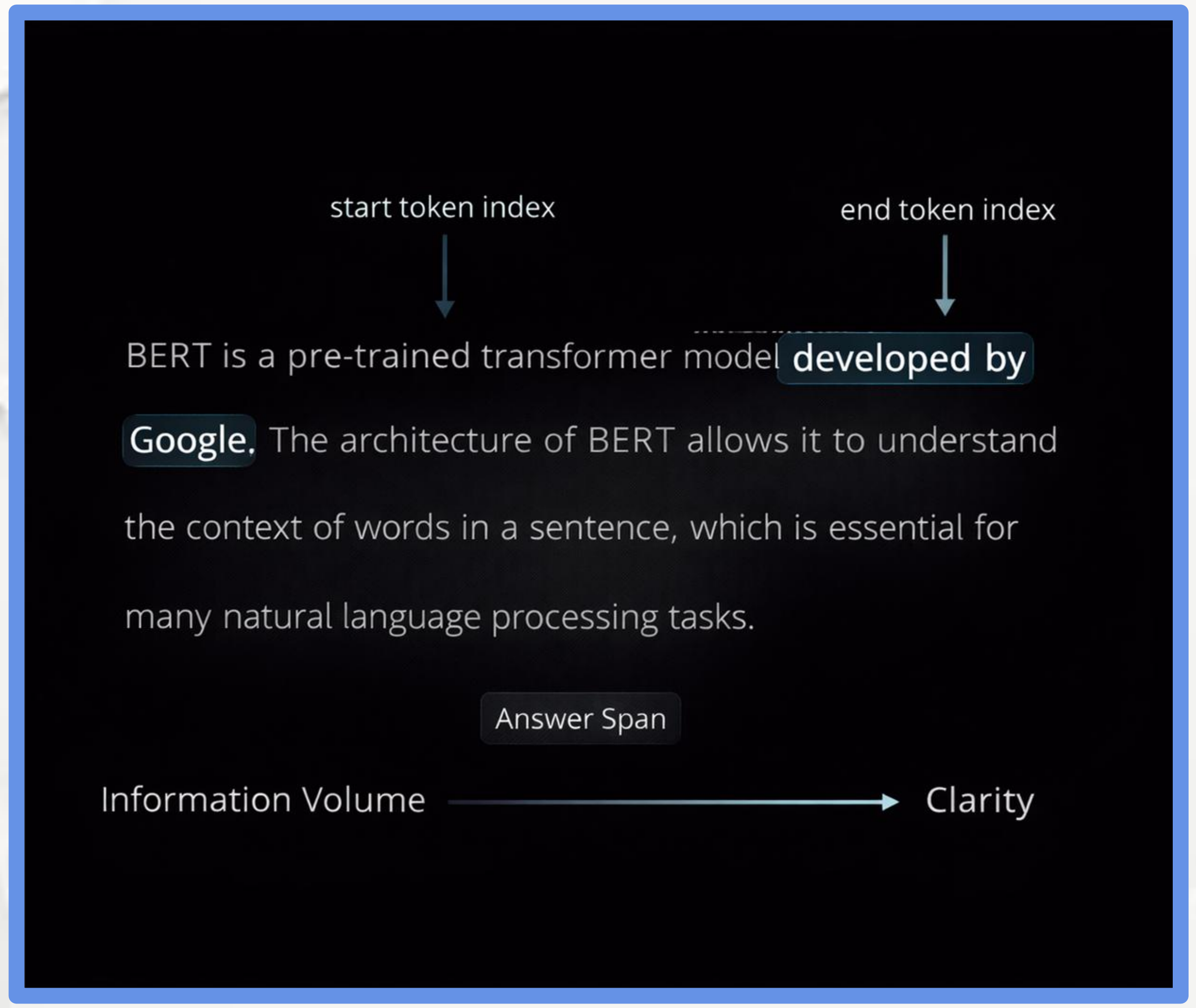
Chapter 5.5 Information Retrieval: Question Answering Systems

Extractive Question Answering

- Answer is a text span
- Start and end token prediction
- High factual precision
- Limited to provided context



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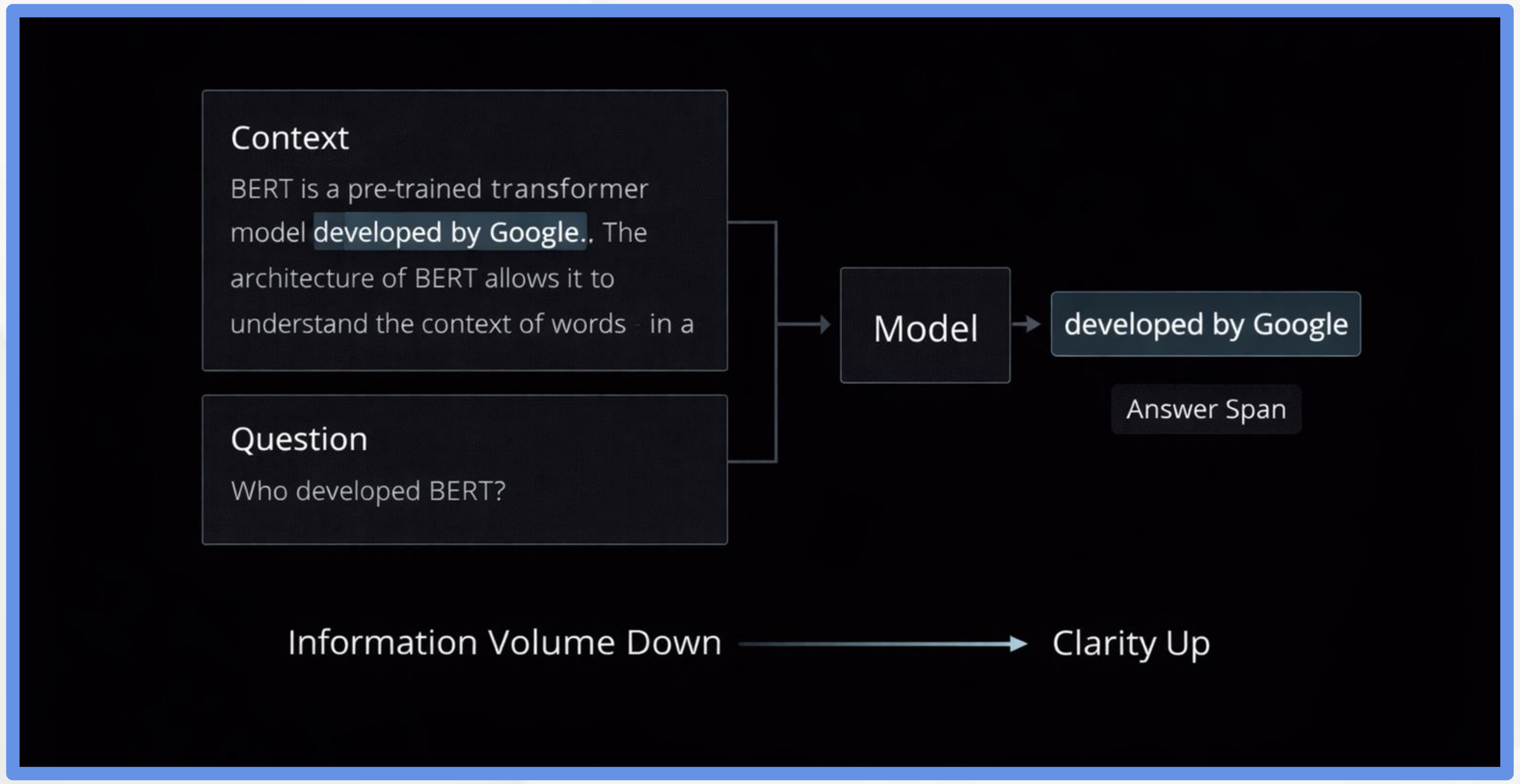
Chapter 5.5 Information Retrieval: Question Answering Systems

The SQuAD Paradigm

- Benchmark dataset for extractive QA
- Context + Question → Span
- Enabled large-scale QA training
- Foundation for enterprise FAQ systems



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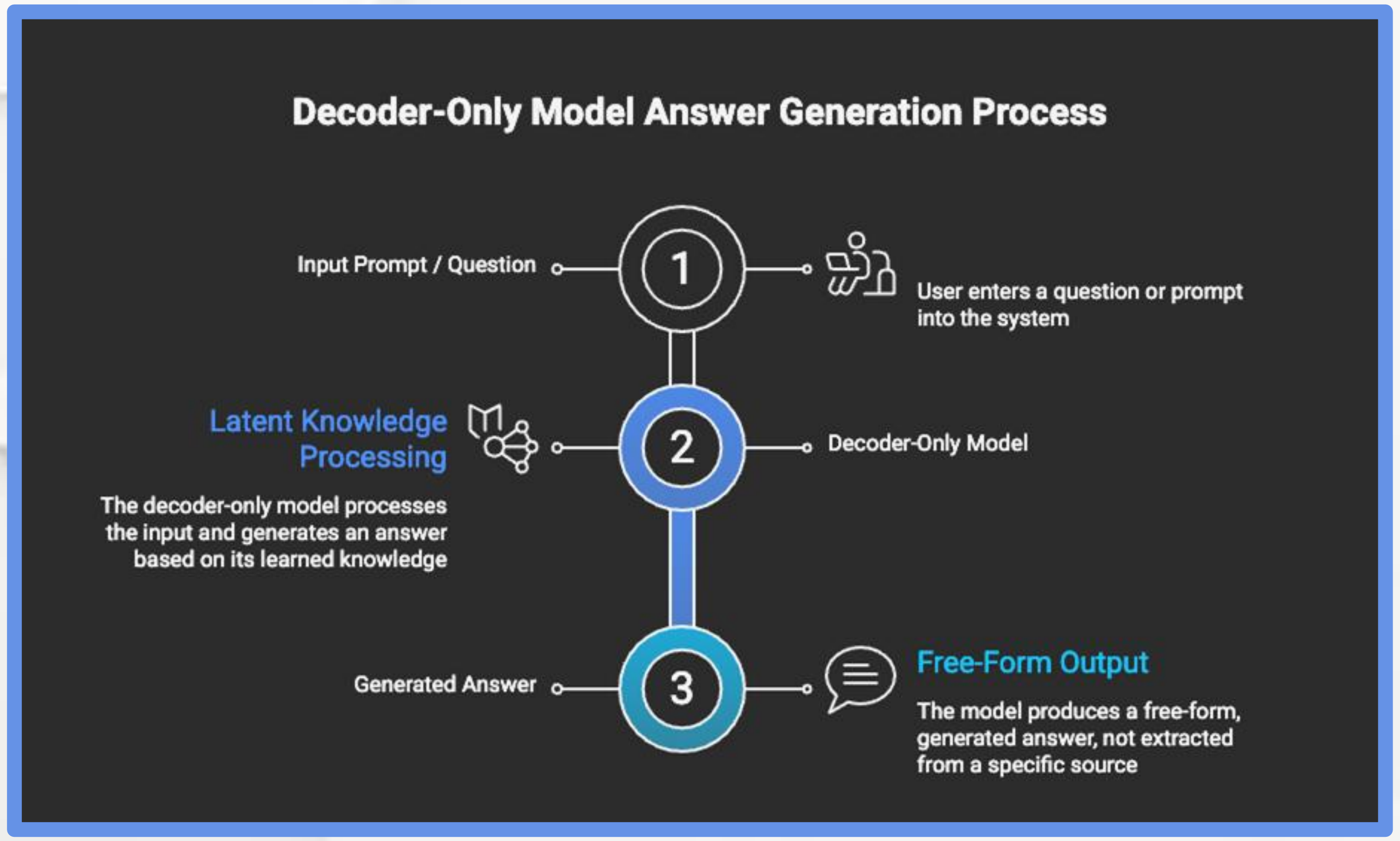
Chapter 5.5 Information Retrieval: Question Answering Systems

Generative Question Answering

- Benchmark dataset for extractive QA
- Context + Question → Span
- Enabled large-scale QA training
- Foundation for enterprise FAQ systems



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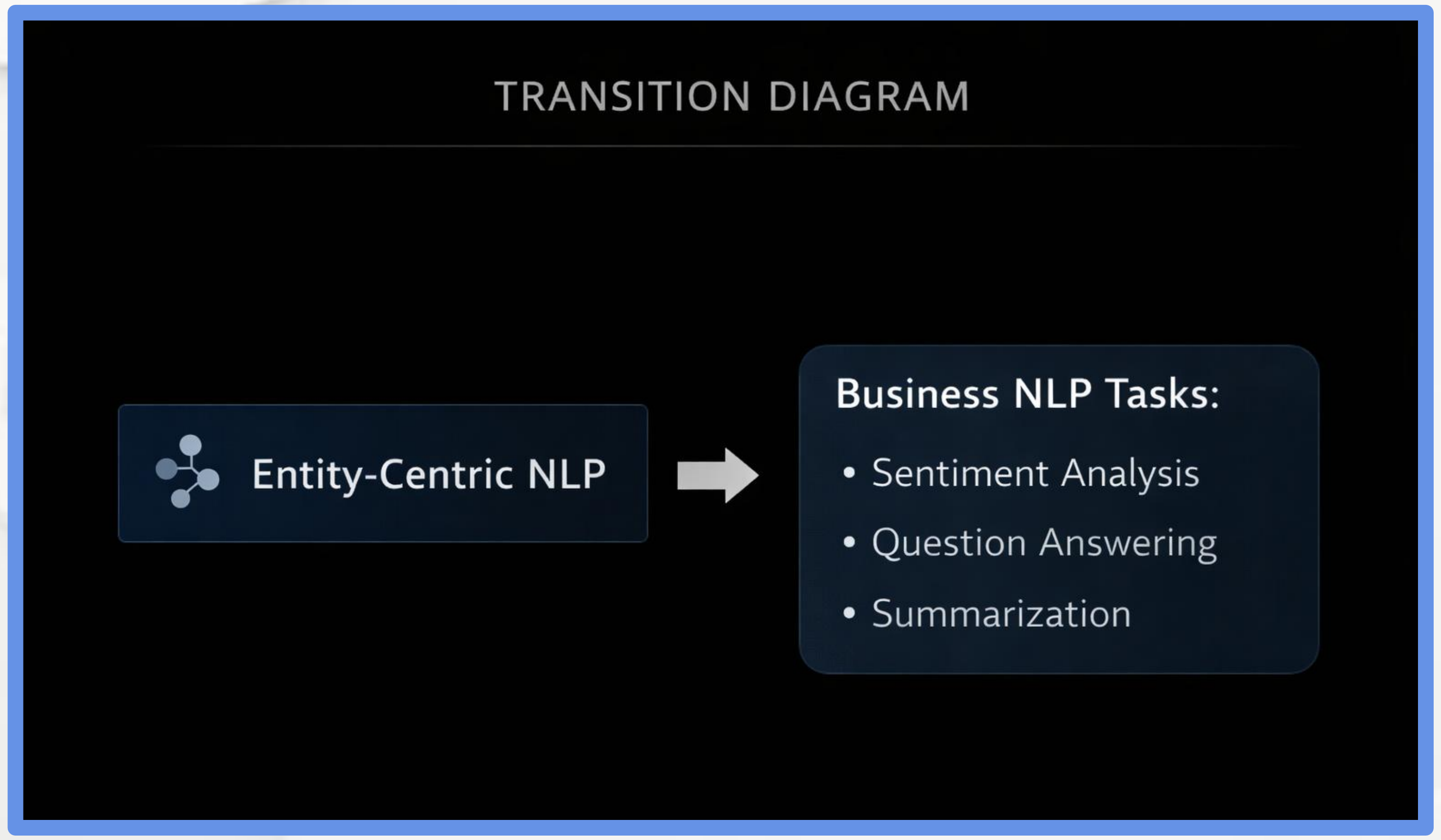
Chapter 5.5 Information Retrieval: Question Answering Systems

Knowledge-Based Question Answering

- Uses structured databases
- Converts language to queries
- Deterministic answers
- High reliability systems



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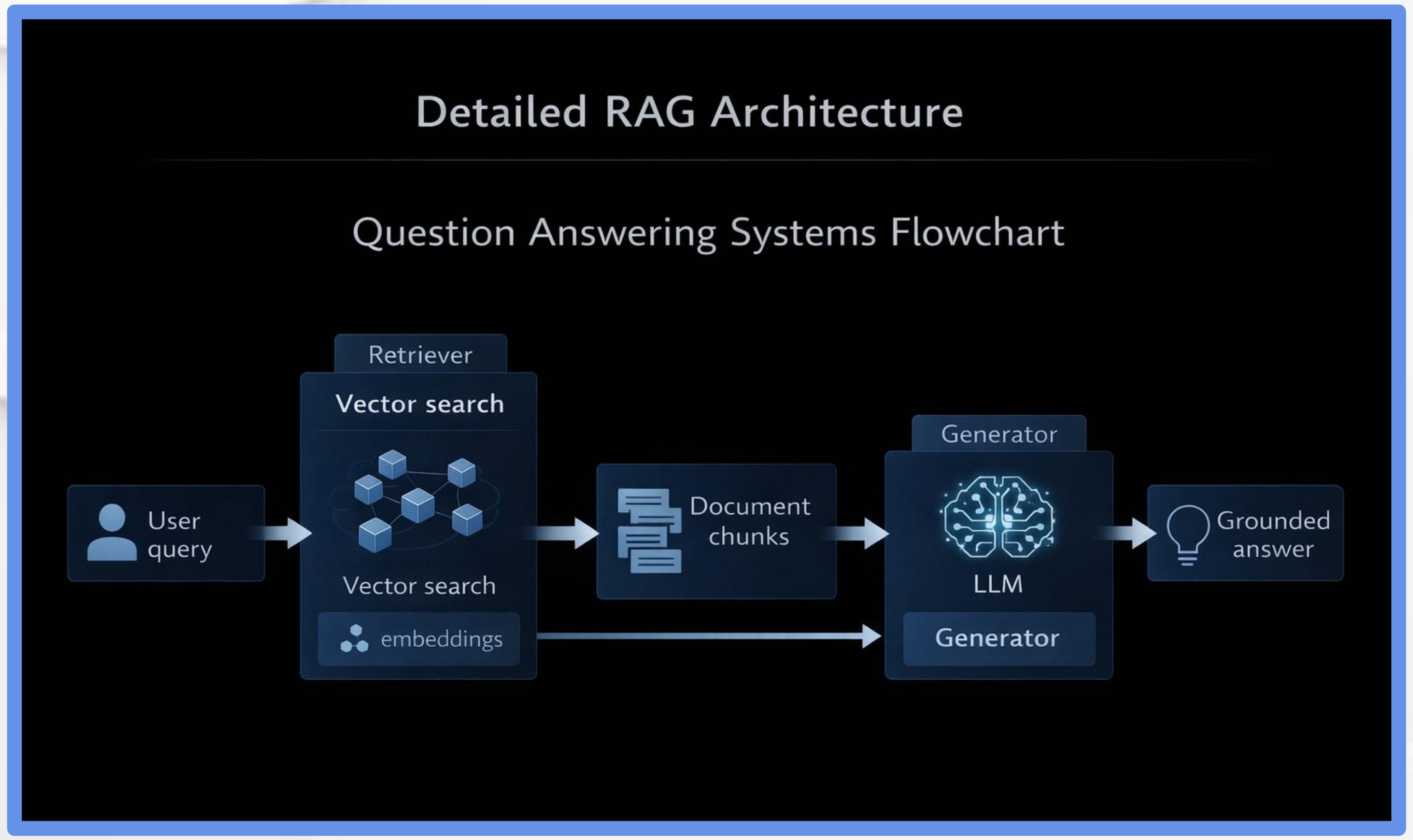
Chapter 5.5 Information Retrieval: Question Answering Systems

Retrieval-Augmented Generation (RAG)

- Industry-standard QA architecture
- Combines retrieval + generation
- Uses private documents
- Reduces hallucinations



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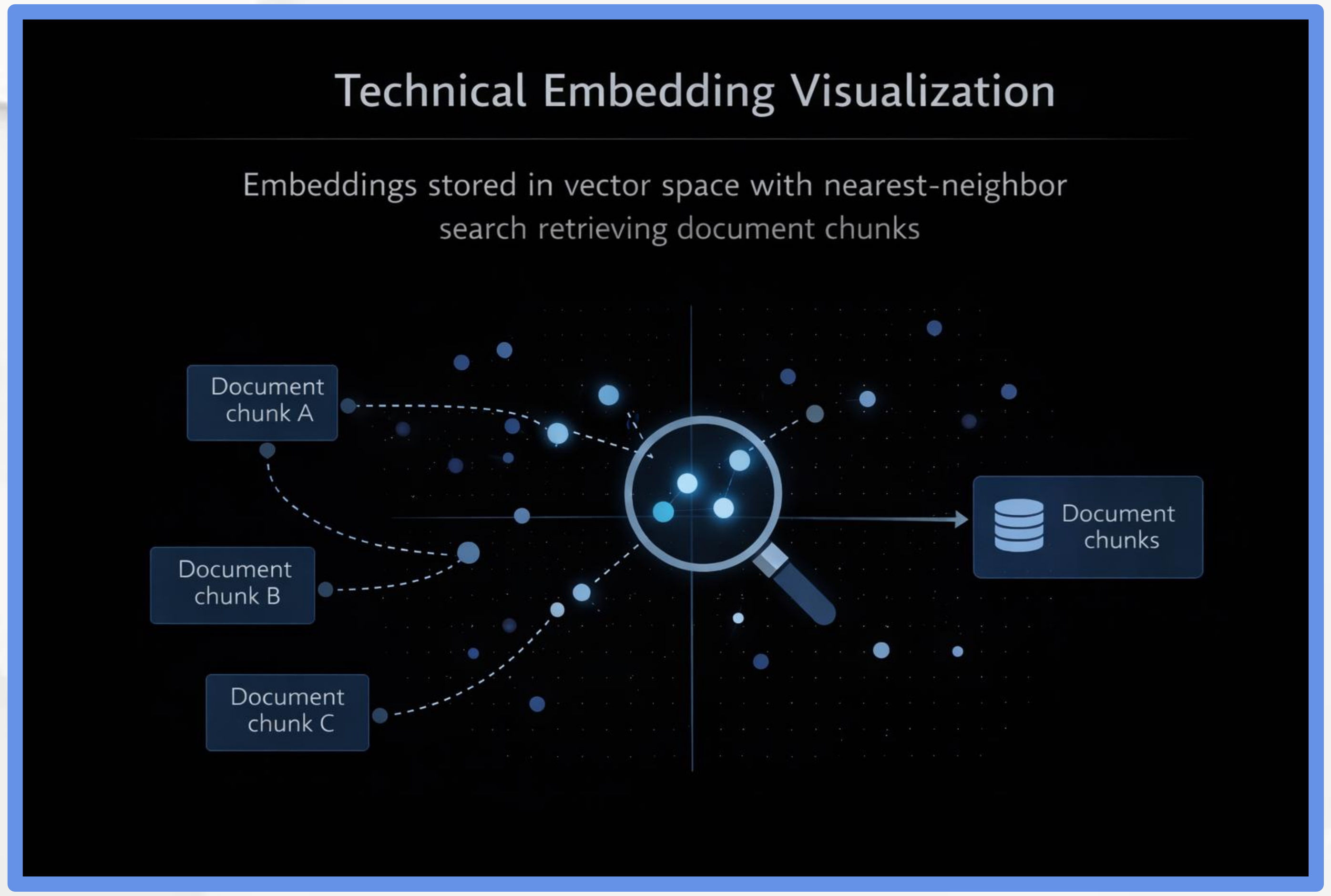
Chapter 5.5 Information Retrieval: Question Answering Systems

Vector Databases in QA

- Exact Match (EM)
- F1-Score (span overlap)
- Faithfulness (RAG-specific)
- Citation correctness



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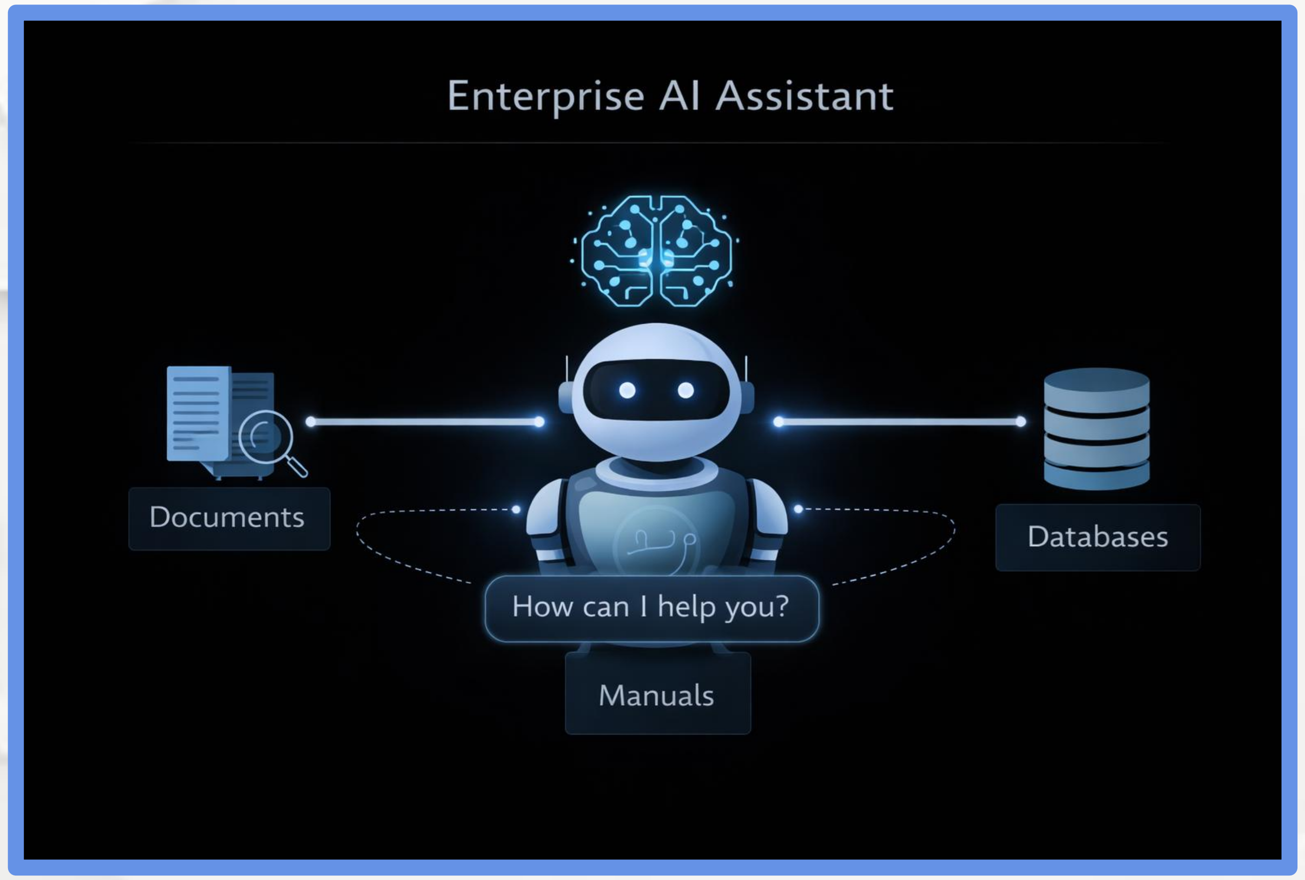
Chapter 5.5 Information Retrieval: Question Answering Systems

Evaluation Metrics for QA

- Customer support automation
- Technical documentation search
- Legal and medical assistants
- Enterprise knowledge access



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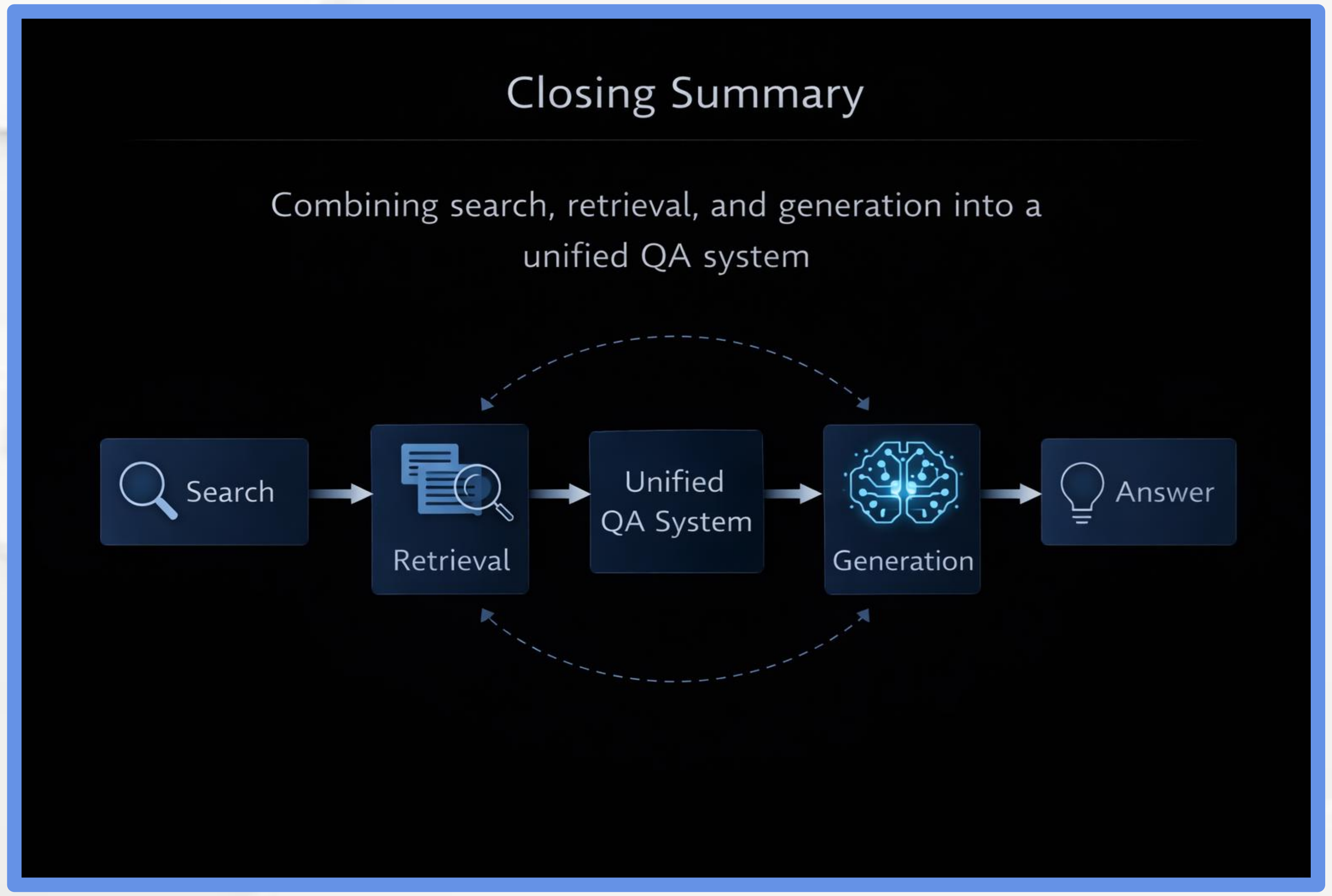
Chapter 5.5 Information Retrieval: Question Answering Systems

Key Takeaways

- Customer support automation
- Technical documentation search
- Legal and medical assistants
- Enterprise knowledge access



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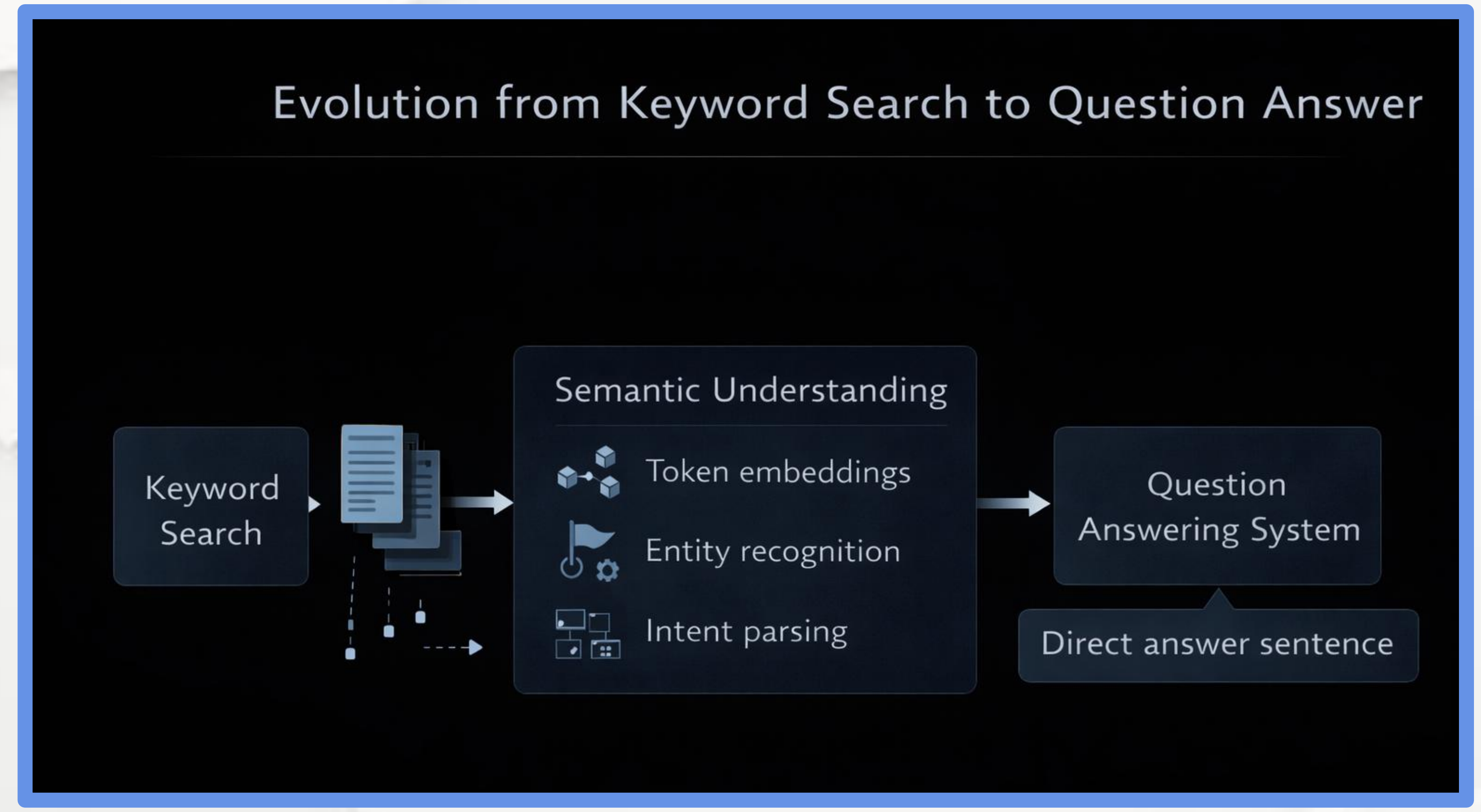
Chapter 5.6 Task Comparison and the Future of Applied NLP

What is Question Answering?

- Customer support automation
- Technical documentation search
- Legal and medical assistants
- Enterprise knowledge access



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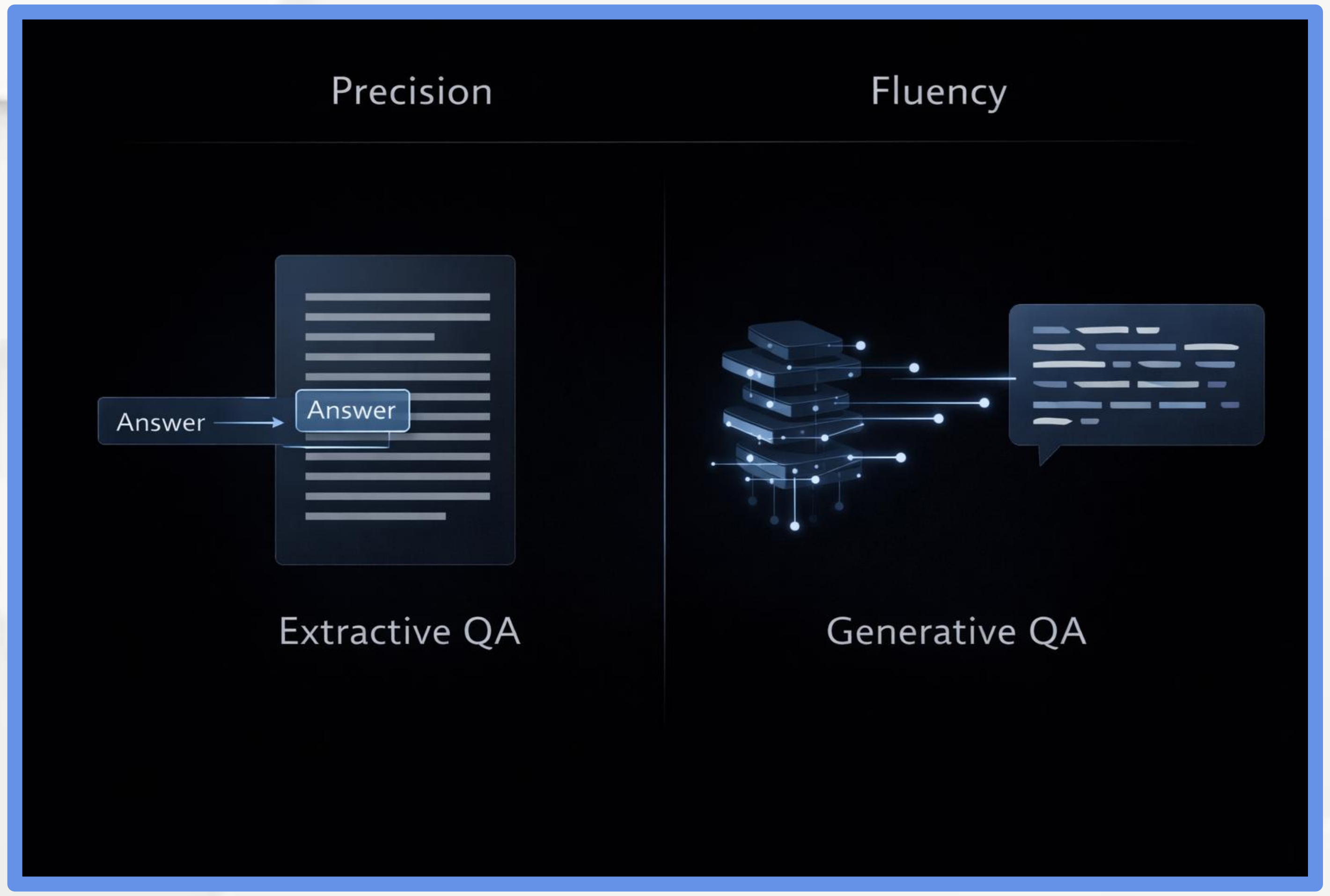
Chapter 5.6 Task Comparison and the Future of Applied NLP

Extractive vs Generative QA

QA systems operate under two main paradigms: extracting exact answers from text or generating new answers using language models.

- Extractive: answer is a text span
- High factual precision
- Used in legal and technical domains
- Generative: produces new text
- More fluent, higher hallucination risk



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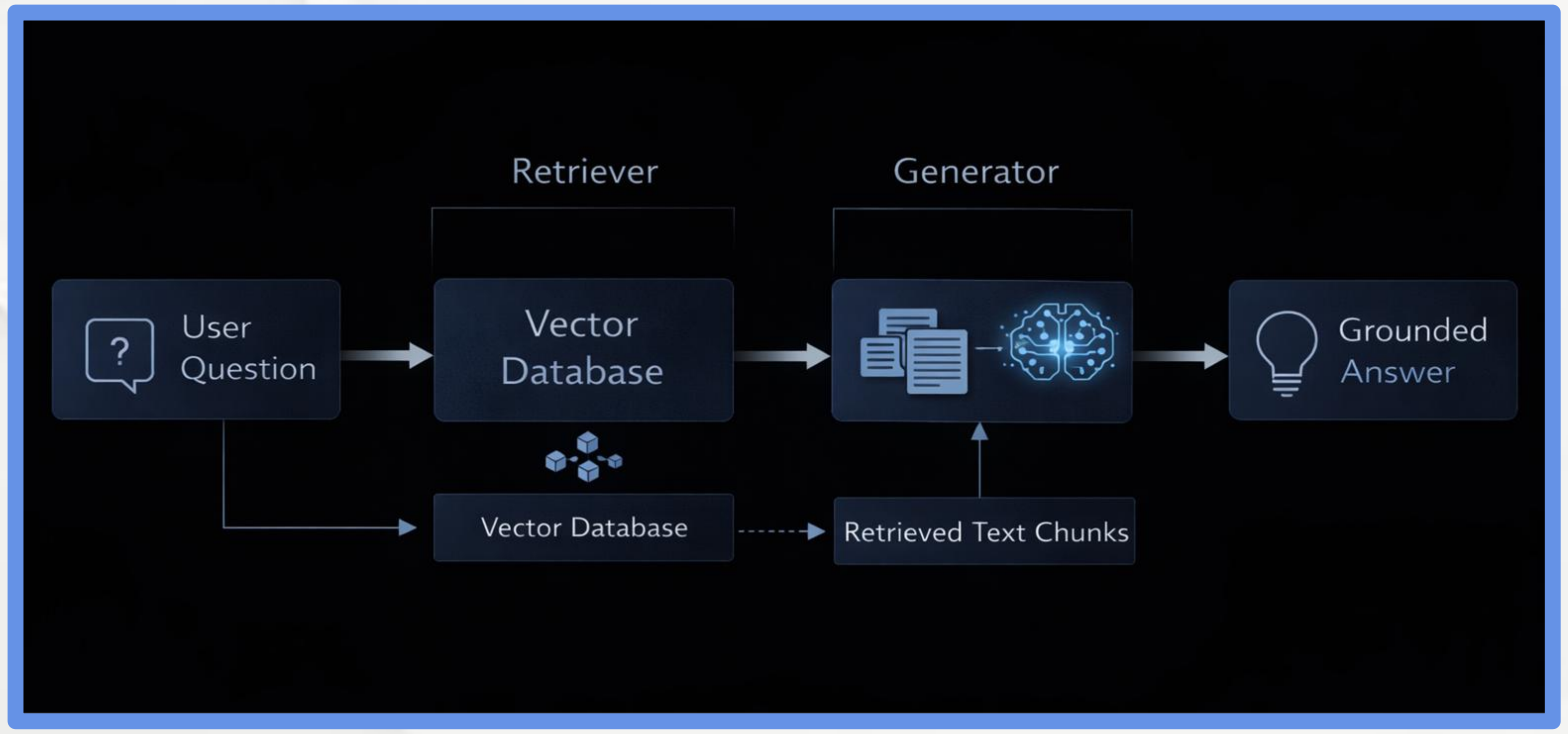
Chapter 5.6 Task Comparison and the Future of Applied NLP

Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) combines document retrieval with generative models to produce answers grounded in real data.

- Industry standard for QA
- Uses vector search
- Injects retrieved text into prompts
- Reduces hallucinations
- Enables private data QA



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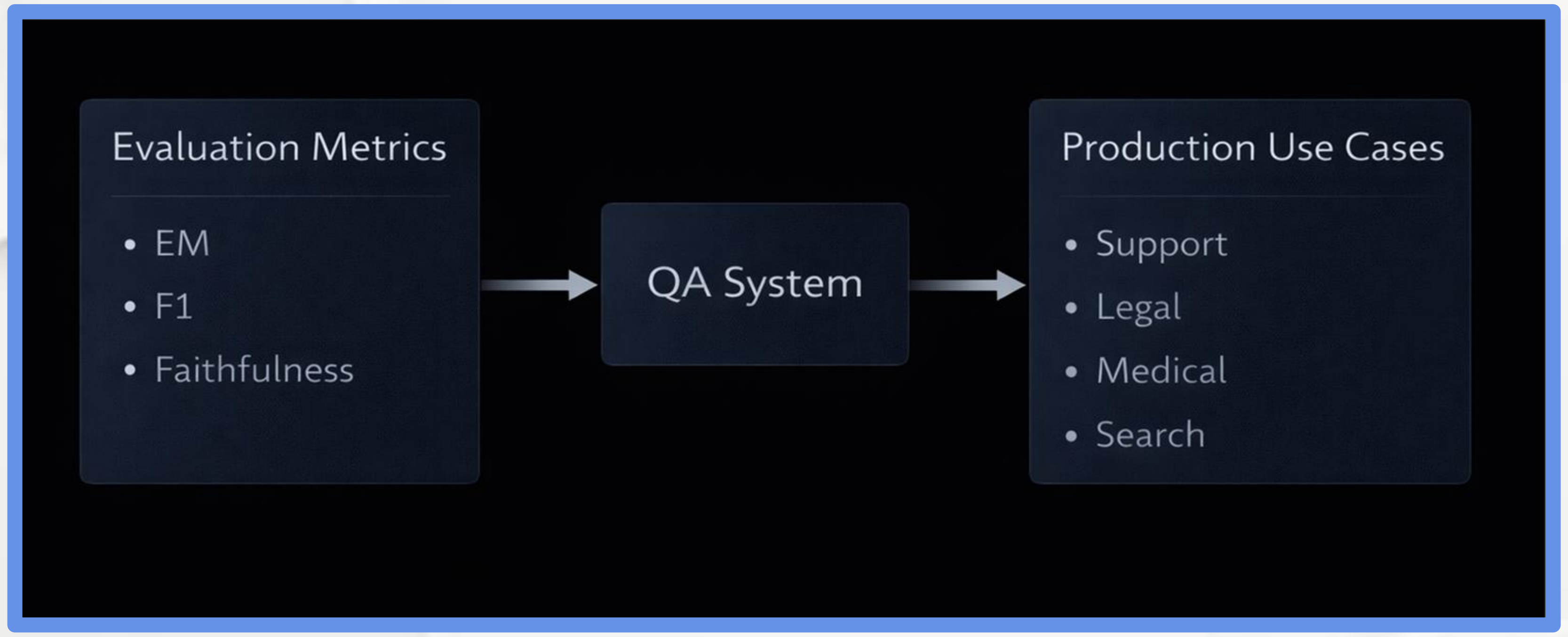
Chapter 5.6 Task Comparison and the Future of Applied NLP

Evaluation and Production Use

Reliable QA systems must be evaluated not only for correctness, but also for trust and factual grounding.

- Exact Match (EM)
- F1-score (span overlap)
- Faithfulness (RAG answers)
- Enterprise QA systems
- Trust and explainability



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